Linear multidimensional regression with interactive fixed-effects *

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Abstract

This paper studies a linear and additively separable model for multidimensional panel data of three or more dimensions with unobserved interactive fixed effects. Two approaches are considered to account for these unobserved interactive fixed-effects when estimating coefficients on the observed covariates. First, the model is embedded within the standard two-dimensional panel framework and restrictions are formed under which the factor structure methods in Bai (2009) lead to consistent estimation of model parameters, but at slow rates of convergence. The second approach develops a kernel weighted fixed-effects method that is more robust to the multidimensional nature of the problem and can achieve the parametric rate of consistency under certain conditions. Theoretical results and simulations show some benefits to standard two-dimensional panel methods when the structure of the interactive fixed-effect term is known, but also highlight how the kernel weighted method performs well without knowledge of this structure. The methods are implemented to estimate the demand elasticity for beer.

1 Introduction

Models of multidimensional data – panel data with more than two dimensions – are becoming increasingly popular in econometric analysis as large data sets with a multidimensional structure become available. For example, in gravity models of trade that are repeated over time one may be interested in studying trade patterns between an importer, i, and an exporter, j, that is repeated every period, t. One may also be interested in studying demand elasticities through

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consumption data that may vary by product, i, store, j, with repeated observation over week or fortnight, t.[1] In these examples it is clear that there may exist unobserved characteristics in each dimension that can determine variation across both the dependent and independent variables that needs to be controlled for to avoid issues with endogeneity. For example, this could be shifts in taste preferences that are unobserved by the econometrician, and may effect sales of particular products in certain stores differently over time. Thus far, most analysis has addressed unobserved heterogeneity in the higher-dimensional setting by using a combination of additive scalar fixed-effects. These approaches, however, can only accommodate variation in unobserved heterogeneity over a subset of dimensions with any one of the scalar fixed-effects terms. For example, in the three-dimensional model this approach can only control for variation over ij, it and jt, but not over all ijt. In the face of more complicated relationships that admit multiplicative variation across dimensions, additive fixed-effects cannot sufficiently control for unobserved heterogeneity. This paper develops tools to control for unobserved heterogeneity in the form of interactive fixed-effects in linear models of multidimensional panel data. The methods developed herein can also be applied in the standard two-dimensional setting for potentially improved convergence rates.

To fix ideas consider linear parameter estimation in the following interactive fixed-effects model with three dimensions,

$$Y_{ijt} = X'_{ijt}\beta + \sum_{\ell=1}^{L} \varphi_{i\ell}^{(1)} \varphi_{j\ell}^{(2)} \varphi_{t\ell}^{(3)} + \varepsilon_{ijt}, \tag{1}$$

where all terms in $\sum_{\ell=1}^{L} \varphi_{i\ell}^{(1)} \varphi_{j\ell}^{(2)} \varphi_{t\ell}^{(3)}$ are unobserved and L is bounded and fixed. The object L is often difficult to calculate, see Håstad (1989). However, for purposes of this analysis it is only necessary to know what is called the multilinear rank, defined in Section 2.1. The multilinear rank of a multidimensional array is the set of matrix-ranks of the different matrices the array can be transformed into along its different dimensions. For example, the multilinear rank of the interactive fixed-effects in three dimensions is a vector of three different ranks - the first being the rank when the first dimensions are the rows, and the second and third dimensions are jointly the columns, and so on for the other multilinear rank entries. In the presence of covariates, estimation of interactive fixed-effect rank is still an open area of research in the standard panel setting. Simulations suggest estimation with more than the required components performs well. Reducing the problem to three dimensions is without loss of generality for the methods considered herein. Additive fixed-effects are omitted for brevity but are subsumed by the interactive fixed-effect term or can be removed with a simple within transformation. Let X_{ijt} be arbitrarily correlated with the unobserved interactive fixed-effects term, $\sum_{\ell=1}^{L} \varphi_{i\ell}^{(1)} \varphi_{j\ell}^{(2)} \varphi_{t\ell}^{(3)}$,

[1] A non-exhaustive list of related examples can be found in the introduction of Matyas (2017).

but uncorrelated with the noise term, ε_{ijt}. The challenge to estimating β is isolating variation in X_{ijt} that is not correlated with the interactive fixed-effects term. This paper develops the multidimensional group fixed-effects and kernel weighted transformations to project out this unobserved heterogeneity and also shows settings where standard factor methods work well. The kernel weighted within transformation is a novel contribution. Group fixed-effects from Bonhomme, Lamadon and Manresa (2022), or the within-cluster transformation in Freeman and Weidner (2023) are also possible in this setting.

This paper makes two main contributions to the literature. The first is to show that the three or higher dimensional model can be couched in a standard two-dimensional panel data model and to derive sufficient conditions for consistency using factor model methods from Bai (2009); Moon and Weidner (2015). The second contribution is to introduce kernel weighted fixed-effects methods. The asymptotic results show that under certain conditions the kernel weighted fixed-effects can achieve the parametric rate of convergence, which are yet to be proven in the three dimensional case using existing methods. The simulation results corroborate these theoretical findings and an empirical application that estimates the demand elasticity of beer demonstrates how these methods work in practice.

The novel estimators proposed in this paper can be described as extensions to the usual within transformation. This makes the estimator accessible for practitioners, and easy to implement. Consider additive fixed-effects of the form $a_{ij} + b_{it} + c_{jt}$. These can be projected out using the transformation,

$$\dot{Y}_{ijt} = Y_{ijt} - \bar{Y}_{\cdot jt} - \bar{Y}_{i\cdot t} - \bar{Y}_{ij\cdot} + \bar{Y}_{\cdot\cdot t} + \bar{Y}_{\cdot j\cdot} + \bar{Y}_{i\cdot\cdot} - \bar{Y}_{\cdot\cdot\cdot}, \tag{2}$$

applied equivalently to X_{ijt}, where the bar variables denote the average taken over the "dotted" index for the entire sample. That is, $\bar{Y}_{\cdot jt} := \frac{1}{N_1}\sum_{i=1}^{N_1} Y_{ijt}$, $\bar{Y}_{\cdot\cdot t} := \frac{1}{N_1 N_2}\sum_{i=1}^{N_1}\sum_{j=1}^{N_2} Y_{ijt}$, etc. where means are taken with equal weights over all observations. In the presence of interactive fixed-effects, the simple within transformation demeans each fixed-effect term to leave,

$$\sum_{\ell=1}^{L}\left(\varphi_{i\ell}^{(1)} - \bar{\varphi}_{\ell}^{(1)}\right)\left(\varphi_{j\ell}^{(2)} - \bar{\varphi}_{\ell}^{(2)}\right)\left(\varphi_{t\ell}^{(3)} - \bar{\varphi}_{\ell}^{(3)}\right)$$

which is clearly not controlled for if the φ terms admit heterogeneity across observations. Of course, the simple within transformation was not designed to control for interactions, but this serves as a useful heuristic.

Consider instead an extension of the within transformation that uses weighted means instead of uniform means. With an abuse of notation, consider,

$$\widetilde{Y}_{ijt} = Y_{ijt} - \bar{Y}_{i^*jt} - \bar{Y}_{ij^*t} - \bar{Y}_{ijt^*} + \bar{Y}_{i^*j^*t} + \bar{Y}_{i^*jt^*} + \bar{Y}_{ij^*t^*} - \bar{Y}_{i^*j^*t^*} \tag{3}$$

where the bar variables combined with the star indices denote weighted means for that observation. For example, $\bar{Y}_{i^*jt} = \sum_{i'} w_{i,i'} Y_{i'jt}$ for weights across i' for each i, that need not be symmetric. In turn, the remainder from the interactive fixed-effects term is,

$$\sum_{\ell=1}^{L} \Big(\varphi_{i\ell}^{(1)} - \sum_{i'} w_{i,i'} \varphi_{i'\ell}^{(1)}\Big)\Big(\varphi_{j\ell}^{(2)} - \sum_{i'} w_{j,j'} \varphi_{j'\ell}^{(2)}\Big)\Big(\varphi_{t\ell}^{(3)} - \sum_{i'} w_{t,t'} \varphi_{t'\ell}^{(3)}\Big).$$

Under regularity conditions, the weighted within transformation can project out the interactive fixed-effects if appropriate weights are used. The same can be true if the weighted means are replaced with cluster means for appropriate cluster assignments, similar to a group fixed-effects estimator. Hence, with a relatively small change to how the within transformation is performed, much more general fixed-effects can be considered in the linear model.

The model for interactive fixed-effects has precedent in the standard two-dimensional panel data setting. For instance take the model considered in Bai (2009) and similar to Pesaran (2006),

$$Y_{it} = X_{it}'\beta + \sum_{\ell=1}^{L} \lambda_{i\ell} f_{t\ell} + e_{it}. \tag{4}$$

In that setting, Bai (2009) show that the interactive term $\sum_{\ell=1}^{L} \lambda_{i\ell} f_{t\ell}$ also sufficiently captures variation in additive individual and time effects without the need to specify these separately. For multidimensional applications the problem in (1) can be simply transformed to a two dimensional problem and estimated as (4) directly using the transformed data. Indeed, Section 3.1 displays useful preliminary convergence rates for this approach. However, convergence rates can suffer severely from the over-parametisation implied by (4). Without strong assumptions on the sparsity of the fixed-effects, only a slow rate of convergence can be guaranteed for this approach. However, when this approach is used to construct preliminary estimators of the fixed-effects, the parametric rate of consistency is possible when these are combined with the novel weighted within transformations. Kuersteiner and Prucha (2020) consider a similar dimension specific projection across the time dimension in the presence of spatial and serial correlation, in particular the Helmert transformation used in Supplement D.5.2.

On top of this slow convergence rate for the two dimensional transformation, finite sample bias may also arise when L is large and only a subset of the unobserved heterogeneity parameters are low-dimensional. For small bias in the estimates of β, transforming the multidimensional array to a matrix then estimating (4) requires either: (a) all the multidimensional fixed-effects are low-dimensional; or (b) that a known subset of the multidimensional fixed effects are low-dimensional, which is a strong assumption. Alternatively, whilst the kernel weighted transformations analysed in this paper also require that a subset of the fixed-effect parameters are low-dimensional, the analyst does not need to know which ones are. In this sense, the novel kernel weighted estimator is robust. A concrete example is considered in a simulation exercise, and there is evidence in the empirical application of differences in dimensionality of each fixed-effect.

The beer demand elasticity application uses Dominick's supermarket data for the Chicago area from 1991-1995, where price and quantity vary over product, store, and fortnight. For comparison, instrumental variables analysis is used to estimate elasticities, with the price of barley used as an instrument. IV estimates show demand is strongly negatively elastic (-2.15). However, estimates are very imprecise. This could be because the instrument only varies over one dimension, time, or, e.g., because prices of other inputs are also highly variable so the instrument does not explain much price variation. Estimates from the weighted within transformation also show demand is downward sloping and elastic (-2.83), but with much better precision. Estimates from existing methods are very sensitive to how the data is transformed into a two-dimensional dataset. The estimates from the weighted-within transformation are similar to the own-price elasticities estimated in Hausman, Leonard and Zona (1994).

The technical component of this paper is related to the numerical analysis literature on low-rank approximations of multidimensional arrays. As pointed out in De Silva and Lim (2008), the low-rank approximation problem in the tensor setting is not well-posed, hence poses technical difficulties. See Kolda and Bader (2009) for a summary of the multidimensional array decomposition problem, and Vannieuwenhoven, Vandebril and Meerbergen (2012); Rabanser, Shchur and Günnemann (2017). Therefore, it is necessary to innovate on this tensor low-rank problem to find appropriate analytical results. To this end, this paper utilises well-posed components of two-dimensional sub-problems for use in nuisance parameter applications. This application has the advantage that they only require the fixed-effects to be projected out at a sufficient asymptotic rate, and do not attempt to directly solve the low-rank tensor problem. [2]

Under sufficient regularity conditions, the methods considered in this paper may also control for variation from arbitrary smooth functions of the low dimensional fixed-effects. Similar to that considered in Zeleneev (2020) and Freeman and Weidner (2023), the functional representation of model (1) could be,

$$Y_{ijt} = X'_{ijt}\beta + h(\varphi_i^{(1)}, \varphi_j^{(2)}, \varphi_t^{(3)}) + \varepsilon_{ijt},$$

for vector-valued $\varphi_i^{(1)}$, $\varphi_j^{(2)}$ and $\varphi_t^{(3)}$. The setting considered in Zeleneev (2020) requires that the transformation is non-smooth, and it is not trivial to see that a "within-type" transformation will sufficiently project out this type of heterogeneity. Formal analysis of this general model are left for future work. Convergence rates for β estimation are also likely to be slower in the functional setting.

[2] Elden and Savas (2011), along with related papers, suggest a reformulation of the low multilinear rank problem that may have promising applications in econometrics.

The proposed methods can also alleviate convergence issues in certain two-dimensional panel data models when N and T are not proportional. Much of the interactive fixed effects literature requires that the ratio $N/T \to \rho \in (0, \infty)$, that is, that N, T grow at the same rate. Consider splitting the larger of the N or T dimension into sub-dimensions. For example, if $N >> T$, then split N into as many sub-dimensions as necessary to make the dimensions proportional. If $N = O(T^\alpha)$ for $\alpha \in (1, M)$, then split N into roughly α equal parts. For example, if $\alpha = 2$,

$$\sum_{\ell=1}^{L} \lambda_{ij\ell} f_{t\ell} = \sum_{\ell=1}^{L} \left(\sum_{\ell'=1}^{L'} \lambda_{i\ell'\ell}^{(1)} \lambda_{j\ell'\ell}^{(2)} \right) f_{t\ell},$$

where $\lambda_{i\ell'\ell}^{(1)}$ and $\lambda_{j\ell'\ell}^{(2)}$ denote the superimposed fixed-effect for the new sub-dimensions. If there is sufficient sparsity in the first dimension fixed-effect parameter, $\lambda_{ij\ell}$, then L' can be bounded and fixed, and the interactive fixed-effect term is synonymous with the multidimensional interactive fixed-effects in (1). This is of course an additional sparsity assumption on the fixed-effects, but may serve as a useful practical solution to settings with vastly different N and T.

The paper is organised as follows: Section 2 introduces the model; Section 3 details the estimators and associated assumptions with convergence results; Section 4 discusses the convergence results and some practical considerations; Section 5 displays the simulation results; Section 6 shows the beer demand estimation empirical application; and Section 7 concludes.

2 Model

Let β^0 denote the true parameter value for the slope coefficients. The model in full dimensional generality is, [3]

$$\boldsymbol{Y} = \sum_{k=1}^{K} \boldsymbol{X}_k \beta_k^0 + \boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon}, \tag{5}$$

where $\boldsymbol{Y}, \boldsymbol{X}_k, \boldsymbol{\varepsilon} \in \mathbb{R}^{N_1 \times N_2 \times \cdots \times N_d}$. $\boldsymbol{\mathcal{A}} = \sum_{\ell=1}^{L} \varphi_\ell^{(1)} \circ \cdots \circ \varphi_\ell^{(d)}$ where $\varphi_\ell^{(n)} \in \mathbb{R}^{N_n}$ for each $n = 1, \ldots, d$ and "$\circ$" is the outer product. L is bounded and fixed. $\boldsymbol{\varepsilon}$ is a noise term uncorrelated with all $\boldsymbol{X}_k$ and all unobserved fixed-effects terms. Take $i_n \in \{1, \ldots, N_n\}$ for all $n \in \{1, \ldots, d\}$ as the dimension specific index, where N_n is the sample size of dimension n. The regressors $\boldsymbol{X}_k$ may be arbitrarily correlated with $\boldsymbol{\mathcal{A}}$. Throughout this paper all dimensions are considered to

[3] For example, in index notation this model can be written as,

$$Y_{i_1,i_2,\ldots,i_d} = \sum_{k=1}^{K} X_{i_1,i_2,\ldots,i_d;k} \beta_k^0 + \mathcal{A}_{i_1,i_2,\ldots,i_d} + \varepsilon_{i_1,i_2,\ldots,i_d}$$

with $\mathcal{A}_{i_1,i_2,\ldots,i_d} = \sum_{\ell=1}^{L} \varphi_{i_1\ell}^{(1)} \cdots \varphi_{i_d\ell}^{(d)}$.

grow asymptotically, that is $N_n \to \infty$ for all n, however, for the asymptotic theory only a subset of two dimensions need to grow asymptotically.

Model (5) can be seen as a natural extension of the Bai (2009) model to three (or more) dimensions with $\boldsymbol{\mathcal{A}}$ interpreted as a "higher-dimensional" factor stucture. Similar to this strand of the literature, all terms in $\boldsymbol{\mathcal{A}}$ are considered fixed nuisance parameters. There are potentially many extensions to the factor model setting in Bai (2009) to the higher dimension case. This paper starts with what seems the most natural extension.

$\mathcal{A}$ may also incorporate additive fixed effects that vary in any strict subset of the dimensions. For example, in the three dimensional setting one may want to control for the additive terms, $a_{ij}+b_{it}+c_{jt}$. These can be controlled for using $L = \min\{N_1, N_2\}+\min\{N_1, N_3\}+\min\{N_2, N_3\}$, with the first $\min\{N_1, N_2\}$ terms $\sum_{\ell=1}^{\min\{N_1,N_2\}} \varphi_{i\ell}^{(1)}\varphi_{j\ell}^{(2)} = a_{ij}$ by setting $\varphi_{t\ell}^{(3)} = 1$ for $\ell = 1, \dots, \min\{N_1, N_2\}$, and so on for the b_{it} and c_{jt}. These could also be controlled for directly using the standard within-transformation before considering the model in (5).

This paper comprises of two main modelling approaches. The first is to embed the multidimensional model into a standard panel data model by simply flattening all arrays into matrices. The second approach uses weighted differences across each dimension to reduce each $\varphi^{(n)}$ component of $\mathcal{A}$ separately for each n. For this reason the model assumptions are split out and stated in Section 3 alongside each estimation approach.

2.1 Notation and preliminaries

For a d-order tensor, $\mathbf{A}$, a factor-n flattening, denoted as $\mathbf{A}_{(n)}$, is the rearrangement of the tensor into a matrix with dimension n varying along the rows and the remaining dimensions simultaneously varying over the columns. That is, $\mathbf{A}_{(n)} \in \mathbb{R}^{N_n \times N_{n+1}N_{n+2}\dots N_1 \dots N_{n-1}}$. The Frobenius norm, $\|\cdot\|_F$, of a matrix or tensor is the entry-wise norm, $\|\mathbf{A}\|_F^2 = \sum_{i_1=1}^{N_1} \cdots \sum_{i_d=1}^{N_d} A_{i_1\dots i_d}^2$. The spectral norm, denoted $\|\cdot\|$, is the largest singular value of a matrix. For a d-order tensor, $\mathbf{A}$, the multilinear rank, denoted $\mathbf{r}$, is a vector of matrix ranks after factor-n flattening in each dimension, with each component of this vector $r_n = rank\big(\mathbf{A}_{(n)}\big)$. Tensor rank, different to multilinear rank, is defined as the least number of outer products of vectors to replicate the tensor. That is, for tensor $\mathbf{A}$ and vectors $u_\ell^{(n)} \in \mathbb{R}^{N_n}$, tensor rank is the smallest L such that $\mathbf{A} = \sum_{\ell=1}^{L} u_\ell^{(1)} \circ \cdots \circ u_\ell^{(d)}$, where $\circ$ is the outer product of a vector. The notation $a \lesssim b$ means the asymptotic order of a is bounded by the asymptotic order of b.

The n-mode product between a tensor $\mathbf{A}$ and matrix B is denoted $\mathbf{A} \times_n B$ and has elements

$$(\mathbf{A} \times_n B)_{i_1,\dots,j,\dots,i_d} = \sum_{i_n=1}^{N_n} A_{i_1,\dots,i_n,\dots,i_d} B_{j,i_n},$$

which is equivalent to saying the flattening $(\mathbf{A} \times_n B)_{(n)} = B\mathbf{A}_{(n)}$. This can be referred to as "hitting" the tensor $\mathbf{A}$ with matrix B in the n^{th} dimension, though this terminology is only stated to help with understanding of the definition.

The singular value decomposition is used in both estimation approaches in this paper so it is important to understand some of its properties. The singular value decomposition of a matrix, $A \in \mathbb{R}^{N_1 \times N_2}$ is

$$A = U\Sigma V' = \sum_{r=1}^{\min\{N_1,N_2\}} \sigma_r u_r v_r', \tag{6}$$

where U is the matrix of left singular vectors, u_r, V is the matrix of right singular vectors, v_r, and Σ is a diagonal matrix of singular values, σ_r, with values running in descending order down the diagonal. For a rank-r matrix, the first r entries on the diagonal of Σ are strictly positive and the remaining entries are zero.

Take the approximation problem,

$$\min_{A'} \|A - A'\|_F \text{ such that } rank(A') = k. \tag{7}$$

It is well known from the Eckart-Young-Mirsky theorem that the solution to this approximation problem is the first k terms of the singular value decomposition, i.e. $\sum_{r=1}^{k} \sigma_r u_r v_r'$. The Eckart-Young-Mirsky theorem effectively picks out the row and column subspaces that best explain variation in the matrix A as the leading columns of the matrix U, respectively of V. The sum of squared error at the minimiser is thus $\sum_{r=k+1}^{\min\{N_1,N_2\}} \sigma_r^2$. This is commonly called a low-rank approximation and forms the cornerstone for estimation of unobserved heterogeneity in the factor model and interactive fixed-effects models in Bai and Ng (2002); Bai (2009); Moon and Weidner (2015) amongst others.

The Eckart-Young-Mirsky theorem, however, does not extend to the three or higher dimensional setting, see De Silva and Lim (2008) for details. To avoid this complication, the multidimensional problem can be translated to the two-dimensional setting to utilise the Eckart-Young-Mirsky theorem, or the fixed-effects parameters need to be shrunk separately, as is done with the weighted-within transformation.

3 Estimation

This section details the three estimation approaches used. The first subsection details how to apply standard two dimensional estimators to the problem and the assumptions required for consistent estimation. The second and third subsections detail the weighted-within transformation approach and the required assumptions for consistency in those settings.

3.1 Matrix low-rank approximation estimator

This section provides a description of some matrix methods that can be applied directly to the multidimensional model and stipulates the assumptions required for consistency. Kapetanios, Serlenga and Shin (2021) employ a similar approach for three-dimensional arrays in conjunction with the Pesaran (2006) common correlated effects estimator. Babii, Ghysels and Pan (2022) employ a similar matricisation procedure as that detailed below, but are interested in inference on the fixed-effect parameters.

Consider recasting the multidimensional array problem into a two dimensional panel problem by flattening Y and X in the n-th dimension,

$$Y_{(n)} = X'_{(n)}\beta^0 + \varphi^{(n)}\Gamma'_n + \varepsilon_{(n)}$$

where $Y_{(n)}, X_{(n)}, \varepsilon_{(n)} \in \mathbb{R}^{N_n \times \prod_{n' \neq n} N_{n'}}$, $\varphi^{(n)}$ is an $N_n \times r_n$ matrix and Γ_n is an $\prod_{n' \neq n} N_{n'} \times r_n$ matrix that accounts for variation in the remaining $\varphi^{(n')}$ for all $n' \neq n$. The term r_n is indexed by the dimension n because it may vary non-trivially according to the flattened dimension. It should then be apparent that this is exactly the model described in (4), that is, the standard linear model with factor structure unobserved heterogeneity as studied in Bai (2009).

The two-dimensional estimator for a given flattening, n, optimises the following objective function,

$$R(\beta, \widehat{r}_n, n) = \min_{\substack{\varphi^{(n)} \in \mathbb{R}^{N_n \times \widehat{r}_n}, \\ \Gamma_n \in \mathbb{R}^{\prod_{n' \neq n} N_{n'} \times \widehat{r}_n}}} \left\| Y_{(n)} - X'_{(n)}\beta - \varphi^{(n)}\Gamma'_n \right\|_F^2 . \tag{8}$$

Then $\widehat{\beta}^{2D}_{(n)} = \operatorname{argmin}_\beta R(\beta, \widehat{r}_n, n)$ is the slope estimate for the two-dimensional setup. The analyst must choose both the dimension to flatten in, n, and the rank of the estimated interactive fixed-effects term, $\widehat{r}_n$. It is well known that the minimum in (8) is achieved using the leading $\widehat{r}_n$ terms from the singular value decomposition of the error term, $Y_{(n)} - X'_{(n)}\beta$. This gives $\widehat{\varphi}^{(n)}$ as the first $\widehat{r}_n$ columns of $\widehat{U}\widehat{\Sigma}$ and $\widehat{\Gamma}_n$ as the first $\widehat{r}_n$ columns of $\widehat{V}$ where $\widehat{U}$, $\widehat{\Sigma}$ and $\widehat{V}$ are the terms from (6) of the singular value decomposition of $Y_{(n)} - X'_{(n)}\beta$. Because this error term is a function of β, an iteration is naturally required between estimating β and finding the singular value decomposition of the error term. This is a well studied iteration procedure; for convergence details see Bai (2009) or Moon and Weidner (2015).

In the following assumptions let $\widehat{r}_n$ be the estimated number of factors for the (n)-flattening of the regression line when applying the least square methods in (8). Also, let $\mathcal{L} \subset \{1, \ldots, d\}$ be a non-empty subset of the dimensions. In the following, the multilinear rank of $\boldsymbol{\mathcal{A}}$ is restricted such that it is low-rank along at least one of the flattenings.

Assumption 1 (Bounded norms of covariates and exogenous error)**.**

(i). $\left\|X_k\right\|_F = O_p\left(\prod_{n=1}^d \sqrt{N_n}\right)$ *for each* k

(ii). $\left\|\varepsilon_{(n^*)}\right\| = O_p\left(\max\{\sqrt{N_{n^*}}, \prod_{m\neq n^*}\sqrt{N_m}\}\right)$ *for each* $n^* \in \mathcal{L}$

Assumption 2 (Weak exogeneity). $\text{vec}(X_k)'\text{vec}(\varepsilon) = O_p\left(\prod_{n=1}^d \sqrt{N_n}\right)$ *for each* k

Assumption 3 (Low multilinear rank). *For some positive integer, c, $r_{n^*} < c$ for all $n^* \in \mathcal{L}$, where r_n is the n^{th} component of the multilinear rank of $\boldsymbol{\mathcal{A}}$.*

Assumption 4 (Non-singularity). *Let $\sigma_s(A)$ be the s^{th} singular value for a matrix A. Consider linear combinations $\delta_{n^*} \cdot X_{(n^*)} = \sum_k \delta_{n^*,k} X_{(n^*),k}$. For each dimension $n^* \in \mathcal{L}$ that satisfies Assumption 3, then for $K \times 1$ unit vector δ_{n^*},*

$$\min_{\{\delta_{n^*}\in\mathbb{R}^K, \|\delta_{n^*}\|=1\}} \sum_{s=r_{n^*}+\widehat{r}_{n^*}+1}^{\min\{N_{n^*}, \prod_{m\neq n^*} N_m\}} \sigma_s^2\left(\frac{(\delta_{n^*}\cdot X_{(n^*)})}{\prod_n \sqrt{N_n}}\right) > b > 0 \qquad wpa1.$$

Assumptions 1, 2 and 4 are standard regularity assumptions already well established in the literature, e.g. see Moon and Weidner (2015). Assumption 1.(i) ensures that the covariates have bounded norms, for example having bounded second moments. Assumption 1.(ii) allows for some weak correlation across dimensions, see Moon and Weidner (2015), or is otherwise implied if the noise terms are independently distributed with bounded fourth moments, see Latała (2005). Assumption 2 is implied if $X_{i_1,i_2,\ldots,i_d;k}\varepsilon_{i_1,i_2,\ldots,i_d}$ are zero mean, bounded second moment and only admits weak correlation across dimensions for each $k = 1,\ldots,K$. Assumption 4 states that, after factor projection, the set of covariates still collectively admit full-rank variation.

Assumption 3 is new and asserts that there exists at least one flattening of the interactive term, $\boldsymbol{\mathcal{A}}$, that is low-dimensional or simply low-rank. This requires that at least one of the unobserved terms $\varphi^{(n)}$ is low dimensional. Note that not all dimensions must satisfy Assumption 3 for the below result. If the correct dimension is chosen then variation from the interactive term can be sufficiently projected out using the factor model approach. This makes up the statement of the following Proposition.

Proposition 1. *Let $\widehat{\beta}^{2D}_{(n^*)}$ be the estimator from Bai (2009) after first flattening along dimension $n^* \in \mathcal{L}$. If Assumptions 1-4 hold, the subset $\mathcal{L}$ is non-empty, and the estimated number of factors $\widehat{r}_{n^*} \geq r_{n^*}$, then, for each $n^* \in \mathcal{L}$ satisfying Assumption 3,*

$$\left\|\widehat{\beta}^{2D}_{(n^*)} - \beta^0\right\| = O_p\left(\frac{1}{\sqrt{\min\{N_{n^*}, \prod_{n\neq n^*} N_n\}}}\right). \tag{9}$$

Proposition 1 follows directly from Moon and Weidner (2015) since the flattening procedure reduces the problem to the standard linear interactive fixed-effects model. Notice that this result only applies to estimates in the dimension(s) that satisfy the low-rank assumption in Assumption 3. That is, implicit in Proposition 1 is that the analyst has chosen the correct dimension to flatten over when reformulating the problem as a two-dimensional panel. The constraint $\widehat{r}_{n^*} \geq r_{n^*}$ can also be changed to $\widehat{r}_{n^*} \geq c$; however, this is more conservative than required for the statement of the result. This constraint does not require knowledge of r_{n^*}, just that the number of estimated factors is greater than or equal the true number.

The estimation procedure from Proposition 1 can also be augmented to flatten over multiple indices. For instance, the analyst may flatten such that both the rows and columns in the matrix contain multiple indices from the original array. Of course, this augmentation makes Assumption 3 harder to satisfy as it requires multiple parameters to vary in low-dimensional space. To see this take the tensor $\boldsymbol{\mathcal{A}}$ flattened over the first two indices as $\mathcal{A}_{(1,2)} \in \mathbb{R}^{N_1 N_2 \times \prod_{n \notin \{1,2\}} N_n}$. If the parameters $\varphi^{(n)}$ for $n = 3, \ldots, d$ are high-dimensional, Assumption 3 is only satisfied when both $\varphi^{(1)}$ and $\varphi^{(2)}$ and their product space is low-dimensional. Clearly this is more restrictive than requiring only one of the parameter spaces to be low-dimensional. However, flattening along multiple dimensions can improve the convergence rate in Proposition 1 to $O_p\left(\frac{1}{\sqrt{\min\{N_1 N_2, \prod_{n \notin \{1,2\}} N_n\}}}\right)$, so there are benefits if this more restrictive assumption can be made. Further discussion of the matrix method results are relegated to Section 4.1, in particular some avenues to choosing the dimension to flatten over.

3.2 Kernel weighted fixed-effects

Let $\widehat{\varphi}_{i_n}^{(n)} \in \widehat{\Phi}_n$ generically denote a known or estimated proxy for fixed-effect $\varphi_{i_n}^{(n)}$. For an example of when this is known, the analyst may have access to exogenous variables that meaningfully explain unobserved heterogeneity in dimension n. The use of this notation will become clear in the statement of Proposition 2 and in discussion of how to estimate these proxy measures in Section 4.2. Let $\mathcal{W}$ be an ordered set of weight matrices, where the n^{th} item $W_n \in \mathbb{R}^{N_n \times N_n}$ has elements,

$$W_{n,i_n i_n'} := \frac{k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\|\right)}{\sum_{i_n'=1}^{N_n} k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{i_n'}^{(n)}\right\|\right)}, \tag{10}$$

where k is a kernel function, and h_n is a bandwidth parameter. The weighted-within transformation in (3) can be generalised with the following series of n-mode products, $\mathbf{Y} \times_1 M_1 \times_2 M_2 \times_3 \cdots \times_d M_d$, likewise also on each $\mathbf{X}_k$, where $M_n = \mathbb{I}_{N_n} - W_n$ and $\times_n$ is the n-mode product defined in Section 2.1. This projection may not optimise any specific objective function - it is designed to sufficiently project out variation in the fixed-effect term. In this sense, the following theoretical analysis is viewed in the light of an asymptotic bias problem. Let $\widehat{\beta}_{KER,\mathcal{W}}$ be the pooled OLS estimator after using weights W_n in the weighed-within transformation on each observed variable. All technical assumptions and results in this section relate to this estimator.

For the remainder of the paper, in particular Section 3.3 for inference, the method used to estimate weights is to use factor model estimates from the matrix method procedure as proxies. Alternatively, if proxies are estimated directly from the error term $\mathbf{Y} - \sum_{k=1}^{K} \mathbf{X}_k \beta_k$, then there is an iteration between slope estimation and estimation of kernel weights. This can also be optimised over a grid space for β if β is low-dimensional.

Assumption 5 (Kernels)**.** *The kernel function, $k(\cdot)$ is, (i). Non-negative, non-increasing, and integrable (ii). $\int k(u)du = 1$ (iii). $\int u^2 k(u)du < \infty$ (iv). $k(u) \lesssim u^{-\delta}$ for $\delta > 2$.*

Assumption 5.(i)-(iii) are standard kernel function restrictions. Assumption 5.(iv) regularises the kernel function to decay sufficiently quickly. This is a weak condition on the decay, and can be satisfied very easily with, e.g., $k(u) = exp(-u^2)$ or $k(u) = \mathbb{1}\{|u| < 1\}$. For consistency, the sequence $h \to 0$ is considered.

Assumption 6 (Regularity of proxies)**.** *Let $\widehat{\varphi}_{i_n}^{(n)} \in \widehat{\Phi}_n$. Define,*

$$M_n^h\left(\widehat{\varphi}_{i_n}^{(n)}\right) := \sum_{i_n' \neq i_n} \mathbb{1}\left(\widehat{\varphi}_{i_n'}^{(n)} \in B_{he}\left(\widehat{\varphi}_{i_n}^{(n)}\right)\right),$$

where $B_{he}(x)$ is an he-neighbourhood around x. For each dimension n, for any $e > 0$, as $h \to 0$, such that $N_n h \to \infty$,

$$\frac{1}{N_n} \sum_{i_n=1}^{N_n} M_n^h\left(\widehat{\varphi}_{i_n}^{(n)}\right) > 0 \qquad wpa.\ 1. \tag{11}$$

Assumption 6 is a restriction on how the fixed-effects proxy parameters are sampled. Equation (11) ensures that there are order N_n observations such that $W_{n,i_n i_n'}$ do not degenerate to 1 when $i_n = i_n'$. It rules out irregular behaviour in how the proxies are sampled, e.g., if $\widehat{\varphi}_{i_n}^{(n)} = i_n$, then the condition is not satisfied for $h \to 0$. If, e.g., $\widehat{\varphi}_{i_n}^{(n)} = i_n / N_n^{\rho}$, then the condition can be satisfied for $N_n^{\rho} h \to \infty$, which restricts the speed of decay $h \to 0$. The restriction in Assumption 6 on the proxies can be used to show a higher-level condition on the sampling of the kernel function, see Lemma A.1 in the Appendix. The condition in Assumption 6 ensures that, asymptotically, neighbourhoods around each proxy are on average non-empty, and avoids the projection $(\mathbb{I}_{N_n} - W_n)$ degenerating to 0 for too many observations.[4]

Assumption 6 is stated for vector-valued $\widehat{\varphi}_{i_n}^{(n)} \in \widehat{\Phi}_n$, which presents a curse of dimensionality, since the requirement scales poorly with $dim(\widehat{\varphi}_{i_n}^{(n)})$. However, the weighted-within transformation can be implemented iteratively, so the restriction can be relaxed to an element-wise restriction on each component of $\widehat{\varphi}_{i_n}^{(n)} \in \widehat{\Phi}_n$, but leads to a significantly more complicated theory. See

[4] Some observations can degenerate, which amounts to removing observations according to outlier fixed-effects proxies. The condition ensures not too many observations are removed.

Appendix A.3 for a treatment of the iterative estimator, along with proof of consistency. Hence, Assumption 6 is viewed as a simplifying assumption.

Assumption 7 (Regularity conditions). *Let $\check{T}_{i_1,\ldots,i_d}$ be the entries of tensor* $\mathbf{T}$ *after the kernel weighted fixed-effects are differenced out. Then,*

(i). $\left(\frac{1}{\prod_n N_n}\sum_{i_1}\cdots\sum_{i_d}\check{X}_{i_1,\ldots,i_d}\check{X}'_{i_1,\ldots,i_d}\right)$ *converges in probability to a nonrandom positive definite matrix as* $N_1,\ldots,N_d\to\infty$.

(ii). $\frac{1}{\prod_n N_n}\sum_{i_1}\cdots\sum_{i_d}\check{X}_{i_1,\ldots,i_d}\varepsilon_{i_1,\ldots,i_d} = O_p\left(\frac{1}{\sqrt{\prod_n N_n}}\right)$.

Assumption 7.(i) is very similar to Assumption 4 except that here full rank is required after the weighted-within projection rather than the factor projection. Assumption 6 restricts the sampling of the fixed-effects proxies to help justify Assumption 7.(i).[5] Assumption 7.(ii) is an exogeneity condition that requires weak exogeneity in the covariates after the weighted-within transformation, which can be viewed as similar to Assumption 2. This is stricter than Assumption 2 because the noise term ε can foreseeably impact weights if weights are estimated from functionals of a residual term. This limitation is alleviated by, for instance, making sure weights are based on variables extraneous to the regression line, hence independent of ε. Likewise, independence can be achieved with a sample split, such as that proposed in Freeman and Weidner (2023). Here weights could be formed from estimates made in one part of the split, and differenced out with the weighted-within transformation from the other part of the split.

Proposition 2 (Upper bound on kernel weighted estimator). *Let Assumptions 5-7 hold. Let* $\frac{1}{N_{n^*}}\sum_{i_{n^*}}\left\|\varphi^{(n^*)}_{i_{n^*}}-\widehat{\varphi}^{(n^*)}_{i_{n^*}}\right\|^2 = O_p(C_{n^*}^{-2})$ *for* $n^*\in\mathcal{M}'$ *with* $C_{n^*}^{-2}\to 0$, *and* $\frac{1}{N_{n'}}\sum_{i_{n'}}\left\|\varphi^{(n')}_{i_{n'}}-\widehat{\varphi}^{(n')}_{i_{n'}}\right\|^2 = O_p(1)$ *for* $n'\notin\mathcal{M}'$, *where* $\mathcal{M}'$ *is a non-empty subset of dimensions. Then,*

$$\left\|\widehat{\beta}_{KER,\mathcal{W}}-\beta^0\right\| = \sqrt{L}O_p\left(\prod_{n^*\in\mathcal{M}'}O_p\left(C_{n^*}^{-1}\right)+O_p\left(h_{n^*}\right)\right)+O_p\left(\prod_{n=1}^{d}\frac{1}{\sqrt{N_n}}\right).$$

For $h_n\lesssim O(C_n^{-1})$ *this reduces to*

$$\left\|\widehat{\beta}_{KER,\mathcal{W}}-\beta^0\right\| = \sqrt{L}O_p\left(\prod_{n^*\in\mathcal{M}'}O_p\left(C_{n^*}^{-1}\right)\right)+O_p\left(\prod_{n=1}^{d}\frac{1}{\sqrt{N_n}}\right).$$

Proposition 2 shows that the convergence rate for the kernel estimator is bounded by the convergence rate of the proxy estimates. That is, as long as the bandwidth parameter approaches zero sufficiently fast, the kernel estimator converges at a rate no worse than the convergence of

[5] It does not imply Assumption 7, but does weaken some conditions for it to hold.

the proxies when proxies are estimations of the true parameter values at or slower than $\sqrt{N_n}$-convergence. This is expected and also a good result that the kernel method does not hinder the convergence rate from these proxies. Discussions in Section 4.2 suggest $O_p(C_{n^*}^{-1})$ can be $O_p\big(1/\sqrt{N_{n^*}}\big)$ under regularity conditions imposed in Bai (2009). This shows the parametric rate is attainable if $\mathcal{M}' = \{1, \ldots, d\}$ and L is fixed and bounded.

Note that implicit in the statement of Proposition 2 is that the correct multilinear rank of the tensor of interactive fixed-effects is known, at least for the dimensions $\mathcal{M}'$. This can likely be relaxed to the case where the upper bound on the multilinear rank is known, but this is left for further research. This would follow from the results in Moon and Weidner (2015), also used in Proposition 1.

3.2.1 Optimal kernel weighted fixed-effects

In the previous section, the proposed estimator was designed to sufficiently project out interactive fixed-effects, without explicitly optimising an objective function. As a diversion, consider an estimator that does optimise an objective function with respect to a weighted fixed-effects model, but adds significant complexity to the technical analysis. This estimator is considered for discussion, and no formal results are derived. Let Δ be a d-list of $\bigtimes_{n=1}^{d} N_n$ tensors $\delta_n \in \mathbb{R}^{N_1 \times \cdots \times N_d}$. For a given set of proxy measures and kernel function, the kernel weighted fixed-effects estimator optimises

$$S(\beta, \mathcal{W}) = \min_{\delta \in \Delta} \left\| \mathbf{Y} - \sum_{k=1}^{K} \mathbf{X}_k \beta_k - \sum_{n=1}^{d} \delta_n \times_n W_n' \right\|_F^2. \tag{12}$$

Then, $\widehat{\beta}_{KEROpt,\mathcal{W}} := \operatorname{argmin}_{\beta \in \mathbb{R}^K} S(\beta, \mathcal{W})$, where $KEROpt$ stands for kernel optimum. The notation $\times_n$ is the n-mode product defined in Section 2.1. Each of the terms in $\sum_{n=1}^{d} \delta_n \times_n W_n'$ can be interpreted as weighted fixed-effects. Take M_n to be an $N_n \times N_n$ matrix defined as $\mathbb{I}_{N_n} - W_n(W_n'W_n)^{\dagger}W_n'$, where † is the Moore-Penrose generalised inverse. Then, the estimator can be formed by pooled OLS after the following series of n-mode products, $\mathbf{Y} \times_1 M_1 \times_2 M_2 \times_3 \cdots \times_d M_d$, likewise also on each $\mathbf{X}_k$. This sequentially differences out the weighted means from each dimension separately. Then the Frisch-Waugh-Lovell theorem straightforwardly applies.

Projection of these weighted fixed-effects is equivalent to using the weighted-within transformation in (3), similar to the previous section, but with different weights. To obtain the kernel optimum estimator, input weights $W_n(W_n'W_n)^{\dagger}W_n'$ to the weighted-within estimator. Using W_n as per the earlier $\widehat{\beta}_{KER,\mathcal{W}}$ estimator normalises weights to $[0, 1]$, which turns out to make technical proofs much simpler. It is expected, however, that the $\widehat{\beta}_{KEROpt,\mathcal{W}}$ estimator will have similar asymptotic properties.

3.3 Asymptotic Distribution

This section establishes an inference procedure for the kernel weighted fixed-effects model. The procedures here do not translate to the matrix low-rank approximation estimator, since convergence rates are too slow for that estimator. Proposition 2 establishes an upper bound on the kernel weighted fixed-effect estimator convergence rate that can be refined to exactly the parametric rate for the fixed-effect asymptotic bias component. To ensure the bias from the fixed-effect term converges sufficiently quickly (i.e. faster than the parametric rate) a further correction is required. Below is an additional orthogonalisation from Freeman and Kristensen (2024) that can achieve this.

Consider the conditional expectation, $\mathbb{E}\left(\boldsymbol{Y}|\boldsymbol{\mathcal{A}}\right) = \Gamma_Y$, $\mathbb{E}\left(\boldsymbol{X}|\boldsymbol{\mathcal{A}}\right) = \Gamma_X$, such that,

$$\boldsymbol{X} = \boldsymbol{\Gamma}_X + \boldsymbol{\eta}, \qquad \boldsymbol{\Gamma}_Y = \boldsymbol{\Gamma}_X \cdot \beta^0 + \boldsymbol{\mathcal{A}}, \qquad \boldsymbol{Y} - \boldsymbol{\Gamma}_Y = (\boldsymbol{X} - \boldsymbol{\Gamma}_X) \cdot \beta^0 + \boldsymbol{\varepsilon}, \tag{13}$$

with $\mathbb{E}\left(\boldsymbol{\eta}|\boldsymbol{\mathcal{A}}\right) = 0$, $\mathbb{E}\left(\boldsymbol{\varepsilon}|\boldsymbol{X}, \boldsymbol{\mathcal{A}}\right) = 0$. The display for $\boldsymbol{X}$ is general, in that if $\boldsymbol{X}$ is unrelated to $\boldsymbol{\mathcal{A}}$, $\boldsymbol{\Gamma}_X = 0$. Hence, the common correlated effects display in (13) is only a representation, not an imposed model. If the correlation between $\boldsymbol{X}$ and $\boldsymbol{\mathcal{A}}$ is weak in the sense that the empirical mean of $\boldsymbol{\Gamma}_X^2$ converges to zero at arbitrary rate, then the following inference correction is not required and the convergence result from Proposition 2 can strengthen to a sufficient rate.

β^0 can be estimated as $\widehat{\beta}_{IC}$, where IC stands for inference correction, using the infeasible estimator,

$$\widehat{\beta}_{IC} = \left(\mathrm{vec}_K(\boldsymbol{X} - \boldsymbol{\Gamma}_X)'\mathrm{vec}_K(\boldsymbol{X} - \boldsymbol{\Gamma}_X)\right)^{-1} \left(\mathrm{vec}_K(\boldsymbol{X} - \boldsymbol{\Gamma}_X)'\mathrm{vec}(\boldsymbol{Y} - \boldsymbol{\Gamma}_Y)\right)$$

where $\mathrm{vec}_K(\boldsymbol{X} - \boldsymbol{\Gamma}_X) \in \mathbb{R}^{\prod_n N_n \times K}$ is shorthand for the matrix of vectorised covariates, with each column the vectorised $(\boldsymbol{X}_k - \boldsymbol{\Gamma}_{X_k})$. The vec notation for $(\boldsymbol{Y} - \boldsymbol{\Gamma}_Y)$ is the standard vectorisation. This is so far infeasible because $\boldsymbol{\Gamma}_X$, and $\boldsymbol{\Gamma}_Y$ need to be estimated.

Consider estimates for $\boldsymbol{\Gamma}_X$, and $\boldsymbol{\Gamma}_Y$ as $\widehat{\boldsymbol{\Gamma}}_X$, and $\widehat{\boldsymbol{\Gamma}}_Y$, respectively, and $\widehat{\Omega}_X = \mathrm{vec}_K(\boldsymbol{X} - \widehat{\boldsymbol{\Gamma}}_X)'\mathrm{vec}_K(\boldsymbol{X} - \widehat{\boldsymbol{\Gamma}}_X)$. Then,

$$\begin{aligned}
\widehat{\Omega}_X^{-1}\mathrm{vec}_K(\boldsymbol{X} - \widehat{\boldsymbol{\Gamma}}_X)'\mathrm{vec}(\boldsymbol{Y} - \widehat{\boldsymbol{\Gamma}}_Y) &= \widehat{\Omega}_X^{-1}\mathrm{vec}_K(\widehat{\boldsymbol{\eta}})'\mathrm{vec}(\boldsymbol{X} \cdot \beta^0 + \boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon} - \widehat{\boldsymbol{\Gamma}}_X \cdot \widetilde{\beta} - \widehat{\boldsymbol{\mathcal{A}}}) \\
&= \widehat{\Omega}_X^{-1}\mathrm{vec}_K(\widehat{\boldsymbol{\eta}})'\mathrm{vec}_K(\widehat{\boldsymbol{\eta}}) \cdot \beta^0 \\
&+ \widehat{\Omega}_X^{-1}\mathrm{vec}_K(\widehat{\boldsymbol{\eta}})\mathrm{vec}\left(\widehat{\boldsymbol{\Gamma}}_X \cdot (\beta^0 - \widetilde{\beta}) + \boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}} + \boldsymbol{\varepsilon}\right) \\
&= \beta^0 + \widehat{\Omega}_X^{-1}B + \widehat{\Omega}_X^{-1}\mathrm{vec}_K(\widehat{\boldsymbol{\eta}})'\mathrm{vec}(\boldsymbol{\varepsilon}),
\end{aligned} \tag{14}$$

where $\widetilde{\beta}$, and $\widehat{\boldsymbol{\mathcal{A}}}$ are preliminary estimates of β^0, and $\boldsymbol{\mathcal{A}}$, respectively, used to form the estimate $\widehat{\boldsymbol{\Gamma}}_Y$. B is a bias term that must be sufficiently small order for standard asymptotic normality arguments. Specifically, $\widehat{\Omega}_X^{-1}B = o_p\left(\prod_{n=1} N_n^{-1/2}\right)$ is required. Hence, preliminary convergence rates are required on $(\mathbf{\Gamma}_X - \widehat{\mathbf{\Gamma}}_X)$, $(\beta^0 - \widetilde{\beta})$, and $(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}})$ to ensure bias is bounded appropriately. From Section 3.2, and the proof in Appendix A, with the kernel weighted estimator, $(\beta^0 - \widetilde{\beta})$ can be bounded by $O_p(\prod_n h_n)$ if $N_n^{-1/2} \lesssim h_n$, where h_n is the bandwidth used for dimension n. Likewise, $\Big(\big(\prod_{n=1} N_n^{-1}\big)\mathrm{vec}(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}})'\mathrm{vec}(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}})\Big)^{1/2} = O_p(\prod_n h_n)$. This leaves the bound for $(\mathbf{\Gamma}_X - \widehat{\mathbf{\Gamma}}_X)$, which is incidentally also useful to bound $\mathrm{vec}_K(\widehat{\boldsymbol{\eta}})$.

Split the bias term, $B = B_1 + B_2$, and let $N := \prod_{n=1} N_n$. Then,

$$\begin{aligned} B_1 &:= \mathrm{vec}_K(\widehat{\boldsymbol{\eta}})'\mathrm{vec}(\widehat{\mathbf{\Gamma}}_X \cdot (\beta^0 - \widetilde{\beta})) \\ B_2 &:= \mathrm{vec}_K(\widehat{\boldsymbol{\eta}})'\mathrm{vec}(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}}). \end{aligned}$$

First, $\mathrm{vec}_K(\widehat{\boldsymbol{\eta}})$ appears in both, which can be split into $\mathrm{vec}_K(\widehat{\boldsymbol{\eta}}) = \mathrm{vec}_K(\mathbf{\Gamma}_X - \widehat{\mathbf{\Gamma}}_X) + \mathrm{vec}_K(\boldsymbol{\eta})$. Let $\frac{1}{N}\mathrm{vec}_K(\mathbf{\Gamma}_X - \widehat{\mathbf{\Gamma}}_X)'\mathrm{vec}_K(\mathbf{\Gamma}_X - \widehat{\mathbf{\Gamma}}_X) = O_p(\xi_X^2)\mathbb{1}_K$, where ξ_X is the rate of consistency for the estimator of $\widehat{\mathbf{\Gamma}}_X$. For the two terms, take $(\beta^0 - \widetilde{\beta}) = O_p(\xi_\beta)\iota_K$, and $\frac{1}{N}\mathrm{vec}(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}})'\mathrm{vec}(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}}) = O_p(\xi_{\mathcal{A}}^2)$, where these rates can be taken from Proposition 2. Then,

$$\begin{aligned} \frac{1}{N}B_1 &= \frac{1}{N}\mathrm{vec}_K(\mathbf{\Gamma}_X - \widehat{\mathbf{\Gamma}}_X)'\mathrm{vec}(\widehat{\mathbf{\Gamma}}_X \cdot (\beta^0 - \widetilde{\beta})) + \frac{1}{N}\mathrm{vec}_K(\boldsymbol{\eta})'\mathrm{vec}(\widehat{\mathbf{\Gamma}}_X \cdot (\beta^0 - \widetilde{\beta})) \\ &\leq O_p(\xi_X\xi_\beta)\iota_K \sum_{k=1}^{K} \frac{1}{\sqrt{N}}\big\|\widehat{\mathbf{\Gamma}}_{X_k}\big\|_F + O_p(\xi_\beta)\frac{1}{N}\sum_{k=1}^{K}\mathrm{vec}_K(\boldsymbol{\eta})'\mathrm{vec}(\widehat{\mathbf{\Gamma}}_{X_k}). \end{aligned}$$

Then, if $\boldsymbol{\eta}_{k,i_1\ldots i_d} \perp\!\!\!\perp \widehat{\mathbf{\Gamma}}_{X_{k'},i_1\ldots i_d}$ for all $k, k' = 1, \ldots K$, and $\frac{1}{\sqrt{N}}\big\|\widehat{\mathbf{\Gamma}}_{X,k}\big\|_F$ is bounded,

$$\frac{1}{N}B_1 = O_p(\xi_X\xi_\beta)\iota_K + O_p(\xi_\beta N^{-1/2})\iota_K$$

Likewise, for B_2,

$$\frac{1}{N}B_2 = O_p(\xi_X\xi_{\mathcal{A}})\iota_K + \frac{1}{N}\mathrm{vec}_K(\boldsymbol{\eta})'\mathrm{vec}(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}}),$$

such that if $\boldsymbol{\eta}_{,i_1\ldots i_d} \perp\!\!\!\perp \widehat{\boldsymbol{\mathcal{A}}}_{i_1\ldots i_d}$

$$\frac{1}{N}B_2 = O_p(\xi_X\xi_{\mathcal{A}})\iota_K + B_{21},$$

where $\mathrm{plim}_{N\to\infty}\sqrt{N}B_{21}$ has zero mean, zero variance.

Under a full rank assumption on the transformed covariates, e.g. Assumption 7, then $\widehat{\Omega}_X^{-1} = O_p(1/N)\mathbb{1}_{K\times K}$, such that,

$$\sqrt{N}(\widehat{\beta}_I - \beta^0) = \sqrt{N}\left(O_p(\xi_X\xi_\beta)\iota_K + O_p(\xi_X\xi_{\mathcal{A}})\iota_K + O_p(\xi_\beta N^{-1/2})\iota_K + \widehat{\Omega}_X^{-1}\mathrm{vec}_K(\widehat{\boldsymbol{\eta}})'\mathrm{vec}(\boldsymbol{\varepsilon})\right).$$

If $\xi_X\xi_\beta = o_p(N^{-1/2})$, $\xi_\beta \to 0$, and $\xi_X\xi_{\mathcal{A}} = o_p(N^{-1/2})$, asymptotic bias is of order $o_p(N^{-1/2})$ and under certain conditions asymptotic normality can be established for the final term. Proposition 2 establishes the upper bound on convergence for ξ_β and $\xi_{\mathcal{A}}$ can be $O_p(N^{-1/2})$, such that for any $\xi_X = o_p(1)$, the $o_p(N^{-1/2})$ convergence rate for asymptotic bias is achievable.

Dependence between either $\boldsymbol{\eta}$, respectively $\boldsymbol{\varepsilon}$, and $\widehat{\boldsymbol{\Gamma}}_X$ or $\widehat{\boldsymbol{\mathcal{A}}}$ can occur because the estimator $\widehat{\boldsymbol{\Gamma}}_X$, and/or $\widehat{\boldsymbol{\mathcal{A}}}$, may be functions of $\boldsymbol{\eta}$, respectively $\boldsymbol{\varepsilon}$. To break this dependence, a simple sample splitting procedure can be implemented. Say the data is split into two sections. The first section can be used for estimation of $\widehat{\boldsymbol{\Gamma}}_X$ and $\widehat{\boldsymbol{\mathcal{A}}}$, and the second section can be used for estimation of $\widehat{\beta}_{IC}$. Hence, as long as $\boldsymbol{\varepsilon}$ and $\boldsymbol{\eta}$ are suitably independent across observations, the dependence introduced by the estimating equations is broken. A concrete example of a sample splitting procedure for panel data is outlined in Freeman and Weidner (2023), which is easily transferable to this setting.

Below are some sampling assumptions to achieve asymptotic normality. Stronger independent and identically distributed assumptions are considered first, which are sequentially relaxed for stronger results later in the section.

Assumption 8. *$\{\varepsilon_{i_1\dots i_d}, \eta_{i_1\dots i_d}\}$, are i.i.d. across $i_1 \dots i_d$.*

Assumption 9. $\mathbb{E}\big(\varepsilon^2_{i_1\dots i_d} | \eta_{i_1\dots i_d}\big) = \mathbb{E}\big(\varepsilon^2_{i_1\dots i_d}\big) =: \sigma^2_\varepsilon \leq M < \infty$

Assumption 10. *$\mathbb{E}\left[\eta_{i_1\dots i_d}\eta'_{i_1\dots i_d}\right]$ is non–singular and $E\left[\varepsilon_{i_1\dots i_d} | \eta_{i_1\dots i_d}\right] = 0$.*

Assumption 11. *Let ξ_X, ξ_β, and $\xi_{\mathcal{A}}$ be sequences bounded in probability as $N_n \to \infty$ for each $n = 1, \dots d$. Maintain $N := \prod_{n=1}^d N_n$. As $N_n \to \infty$ for each $n = 1, \dots d$, $\{\xi_X, \xi_\beta\} \to 0$, $\xi_X\xi_\beta = o_p(N^{-1/2})$ and $\xi_X\xi_{\mathcal{A}} = o_p(N^{-1/2})$, where: (i). $\left(\|\widehat{\boldsymbol{\Gamma}}_{X_k} - \boldsymbol{\Gamma}_{X_k}\|_F^2/N\right)^{1/2} = O_P\left(\xi_X\right)$ for each $k = 1, \dots K$, (ii). $\left(\|\widehat{\boldsymbol{\mathcal{A}}} - \boldsymbol{\mathcal{A}}\|_F^2/N\right)^{1/2} = O_P\left(\xi_{\mathcal{A}}\right)$, (iii). $\|\beta - \widetilde{\beta}\| = O_p\left(\xi_\beta\right)$.*

Assumptions 8 and 9 are standard i.i.d. and homoskedasticity assumptions that are relaxed later. Assumption 11.(i) implies that there exists a consistent estimate of $\mathbb{E}\left(\boldsymbol{X}|\boldsymbol{\mathcal{A}}\right)$, with an arbitrary convergence rate. For implementation, the kernel weighted estimator, assuming X follows a pure interactive fixed-effect model can achieve this with regularity conditions on how $\boldsymbol{\mathcal{A}}$ enters the data generating process for X. The convergence rate ξ_X can be arbitrarily slow as long as $\xi_{\mathcal{A}}$ and ξ_β converge to zero sufficiently fast. Hence, in the context of Proposition 2, this is not an overly strong assumption on $\widehat{\boldsymbol{\Gamma}}_{X_k}$. The convergence rates in Assumption 11.(ii)-(iii) can be taken directly from Proposition 2, e.g., $\xi_{\mathcal{A}} = O_p(1/\sqrt{N})$, and $\xi_\beta = O_p(1/\sqrt{N})$ under some additional regularity conditions noted below.

Assumption 11 details a potential reprieve from $\xi_{\mathcal{A}} = O_p(1/\sqrt{N})$, and $\xi_\beta = O_p(1/\sqrt{N})$ if indeed it is possible for ξ_X to converge at a faster rate. For example, to achieve $\xi_{\mathcal{A}} = O_p(1/\sqrt{N})$, and $\xi_\beta = O_p(1/\sqrt{N})$, it is necessary that the bandwidth parameter h_n shrink at the rate $1/\sqrt{N_n}$ for each $n = 1, \dots, d$, faster than $O_p({N_n}^{-1/5})$ usually imposed for nonparametric estimation. Hence, for sufficiently fast convergence rate ξ_X, the bandwidth parameter h_n has slack, and can converge slower for better efficiency.

The next assumption restricts $\boldsymbol{\eta}$ and $\boldsymbol{\varepsilon}$ to be independent of $\widehat{\boldsymbol{\Gamma}}_X$ and $\widehat{\boldsymbol{\mathcal{A}}}$. As mentioned already, a sample splitting device can be used to achieve this independence, given an appropriate independence condition such as Assumption 8, or weak correlations.

Assumption 12. *Let $\widehat{\boldsymbol{\Gamma}}_X$ be the estimate for $\boldsymbol{\Gamma}_X$ and $\widehat{\boldsymbol{\mathcal{A}}}$ the estimate for $\boldsymbol{\mathcal{A}}$ in* (13). *Then, (i). $\boldsymbol{\eta}_{k,i_1\dots i_d} \perp\!\!\!\perp \boldsymbol{\mathcal{A}}_{i_1\dots i_d}, \widehat{\boldsymbol{\mathcal{A}}}_{i_1\dots i_d}$ for all $k = 1,\dots K$; (ii). $\boldsymbol{\eta}_{k,i_1\dots i_d} \perp\!\!\!\perp \widehat{\boldsymbol{\Gamma}}_{X_{k'},i_1\dots i_d}$ for all k, k'; (iii). $\boldsymbol{\varepsilon}_{i_1\dots i_d} \perp\!\!\!\perp \widehat{\boldsymbol{\Gamma}}_{X'_k,i_1\dots i_d}$ for all $k = 1,\dots K$.*

The implications of Assumption 12.(i)-(ii) on the convergence of the inference estimator are already discussed above. Independence between $\boldsymbol{\eta}_{k,i_1\dots i_d}$ and $\boldsymbol{\mathcal{A}}_{i_1\dots i_d}$ in Assumption 12.(i) is made for convenience for a cleaner expression of the term B_{21} above, but can be relaxed under further regularity conditions on the moments of $\boldsymbol{\eta}_{k,i_1\dots i_d}$. Assumption 12.(iii) implies that in (14), $\widehat{\Omega}_X^{-1}\text{vec}_K(\widehat{\boldsymbol{\eta}})'\text{vec}(\boldsymbol{\varepsilon}) = \widehat{\Omega}_X^{-1}\text{vec}_K(\boldsymbol{\eta})'\text{vec}(\boldsymbol{\varepsilon}) + o_p(N^{-1/2})$. Under the assumptions imposed in Proposition 2, with $C_n^{-1} \asymp h_n = O_p(1/\sqrt{N_n})$ for all $n = 1,\dots,d$ and $\mathcal{M} = \{1,\dots,d\}$, and Assumptions 8-12, maintaining that $N := \prod_{n=1} N_n$,

$$\begin{aligned}\sqrt{N}(\widehat{\beta}_I - \beta^0) &= o_p(1) + \left(\frac{1}{N}\widehat{\Omega}_X\right)^{-1}\frac{1}{\sqrt{N}}\text{vec}_K(\boldsymbol{\eta})'\text{vec}(\boldsymbol{\varepsilon}) \\ &= o_p(1) + \left(\mathbb{E}\left[\boldsymbol{\eta}_{i_1\dots i_d}\boldsymbol{\eta}'_{i_1\dots i_d}\right] + o_p(1)\right)^{-1}\frac{1}{\sqrt{N}}\text{vec}_K(\boldsymbol{\eta})'\text{vec}(\boldsymbol{\varepsilon})\end{aligned}$$

Under a central limit theorem for the term $\frac{1}{\sqrt{N}}\text{vec}_K(\boldsymbol{\eta})'\text{vec}(\boldsymbol{\varepsilon})$, the estimator is then asymptotically normal with variance,

$$\mathbb{E}\left[\boldsymbol{\eta}_{i_1\dots i_d}\boldsymbol{\eta}'_{i_1\dots i_d}\right]^{-1}\Omega\mathbb{E}\left[\boldsymbol{\eta}_{i_1\dots i_d}\boldsymbol{\eta}'_{i_1\dots i_d}\right]^{-1}.$$

Under Assumption 8 and 9, $\Omega = \mathbb{E}\left[\boldsymbol{\eta}_{,i_1\dots i_d}\boldsymbol{\eta}'_{,i_1\dots i_d}\right]\sigma_\varepsilon^2$, for $\sigma_\varepsilon^2 := \mathbb{E}(\boldsymbol{\varepsilon}^2_{i_1\dots i_d})$, such that the variance simplifies in the standard way to $\sigma_\varepsilon^2\mathbb{E}\left[\boldsymbol{\eta}_{,i_1\dots i_d}\boldsymbol{\eta}'_{,i_1\dots i_d}\right]^{-1}$. Therefore,

$$\sqrt{N}(\widehat{\beta}_{IC} - \beta^0) \underset{d}{\to} \mathcal{N}\left(0, \sigma_\varepsilon^2\mathbb{E}\left[\boldsymbol{\eta}_{i_1\dots i_d}\boldsymbol{\eta}'_{i_1\dots i_d}\right]^{-1}\right)$$

Below regularity conditions from Bai (2009); Bai and Ng (2002) ensure estimates of the fixed-effects used as proxies in the kernel weights converge at the sufficient rate.

Assumption 13 (Bai (2009) conditions)**.** *Let $N := \prod_{n=1} N_n$. (i). $\mathbb{E}\|X_{i_1\dots i_d}\|^4$ bounded.*

(ii). For each $n = 1,\dots,d$, $\mathbb{E}\|\varphi^{(n)}_{i_n}\|^4$ bounded, and $\frac{1}{N_n}\sum_{i_n=1}^{N_n}\varphi^{(n)}_{i_n}\varphi^{(n)\prime}_{i_n}$ converges to an $r_n \times r_n$ p.d. matrix as $N_n \to \infty$. (iii). $\mathbb{E}\varepsilon_{i_1\dots i_d} = 0$, $\mathbb{E}(\varepsilon_{i_1\dots i_d})^8$ bounded. (iv). $\mathbb{E}\varepsilon_{i_1\dots i_d}\varepsilon_{i'_1\dots i'_d} = \tau_{i_1\dots i_d;i'_1\dots i'_d}$, and $N^{-1}\sum_{i_1\dots i_d}\sum_{i'_1\dots i'_d}|\tau_{i_1\dots i_d;i'_1\dots i'_d}|$ bounded. (v). $\varepsilon_{i_1\dots i_d}$ independent of $X_{i'_1\dots i'_d}$ and $\varphi^{(n)}_{i'_n}$ for all $i_1\dots i_d$ and $i'_1\dots i'_d$, and $n = 1,\dots,d$.

(vi). For $\sigma^{(n)}_{i_n,j_n,k_n,m_n} = cov(\varepsilon_{i_1,\dots,i_n,\dots,i_d}\varepsilon_{i_1,\dots,j_n,\dots,i_d}, \varepsilon_{i_1,\dots,k_n,\dots,i_d}\varepsilon_{i_1,\dots,m_n,\dots,i_d})$, for all $n = 1,\dots,d$;

$$N_n^{-2}\prod_{n'\neq n}N_{n'}^{-1}\sum_{i_n,j_n,k_n,m_n}\sum_{i_{n'},n'\neq n}|\sigma^{(n)}_{i_n,j_n,k_n,m_n}| \leq M < \infty$$

Corollary 1. *If Assumption 13, and assumptions in Proposition 1 hold;* $C_n^{-2} = 1/\min\{N_n, \prod_{n' \neq n} N_{n'}\}$ *for each* $n = 1, \ldots, d$ *if the Bai (2009) estimator is used in the matrix low-rank setting from Section 3.1 separately for each* $n = 1, \ldots, d$.[6]

Theorem 1 (Asymptotic distribution under homoskedasticity)**.** *Let the assumptions in Proposition 2 hold. Additionally, let Assumptions 8-13 hold. Then, for* $N_n \to \infty$, *with* $N_n \lesssim \prod_{n' \neq n} N_{n'}$, *for all* $n = 1, \ldots, d$,

$$\sqrt{N}\big(\widehat{\beta}_{IC} - \beta^0\big) \underset{d}{\to} \mathcal{N}\left(0, \sigma_\varepsilon^2 \mathbb{E}\left[\boldsymbol{\eta}_{i_1 \ldots i_d} \boldsymbol{\eta}'_{i_1 \ldots i_d}\right]^{-1}\right)$$

Consistent estimation of the variance term is possible under the same set of assumptions. Estimate $\widehat{\boldsymbol{\varepsilon}} = \boldsymbol{Y} - \boldsymbol{X} \cdot \widehat{\beta}_{IC} - \widehat{\boldsymbol{\mathcal{A}}}$. Then, $\frac{1}{N} \sum_{i_1 \ldots i_d} \widehat{\varepsilon}^2_{i_1 \ldots i_d} = \sigma_\varepsilon^2 + o_p(1)$ follows from the consistency of $\widehat{\boldsymbol{\Gamma}}_Y$, $\widehat{\boldsymbol{\Gamma}}_X$, and $\widehat{\beta}_{IC}$. Likewise $\mathbb{E}\left[\boldsymbol{\eta}_{i_1 \ldots i_d} \boldsymbol{\eta}'_{i_1 \ldots i_d}\right]$ can be estimated consistency from the sample analog $\frac{1}{N} \sum_{i_1 \ldots i_d} \left(\boldsymbol{X}_{i_1 \ldots i_d} - \widehat{\boldsymbol{\Gamma}}_{X, i_1 \ldots i_d}\right)\left(\boldsymbol{X}_{i_1 \ldots i_d} - \widehat{\boldsymbol{\Gamma}}_{X, i_1 \ldots i_d}\right)'$. A degrees of freedom adjustment can improve finite sample properties of estimated standard errors.

Some less restrictive assumptions on the error term can be used, and would require Lyapunov conditions. Alternatively, a central limit theorem can be assumed, as is common in the interactive fixed-effects literature, for cases that drop the identically distributed assumption.[7]

Assumption 14. $\varepsilon_{i_1 \ldots i_d}$ *are independent across* $i_1 \ldots i_d$. $\eta_{i_1 \ldots i_d}$ *are i.i.d. across* $i_1 \ldots i_d$.

Assumption 15. *(i).* $\sigma^2_{i_1 \ldots i_d} := \mathbb{E}\big(\varepsilon^2_{i_1 \ldots i_d} | \boldsymbol{\eta}\big)$ *is bounded.*

(ii). For nonrandom positive definite Σ,

$$\operatorname*{plim}_{N \to \infty} \frac{1}{N} \sum_{i_1 \ldots i_d} \sigma^2_{i_1 \ldots i_d} \eta_{i_1 \ldots i_d} \eta'_{i_1 \ldots i_d} = \Sigma,$$

$$\frac{1}{\sqrt{N}} \sum_{i_1 \ldots i_d} \eta_{i_1 \ldots i_d} \varepsilon_{i_1 \ldots i_d} \xrightarrow{d} \mathcal{N}(0, \Sigma).$$

Assumption 15 can be amended to restrict how much heteroskedasticity and correlation across units is allowed. Then,

$$\sqrt{N}\big(\widehat{\beta}_{IC} - \beta^0\big) \xrightarrow{d} \mathcal{N}(0, \Omega_X^{-1}\, \Sigma\, \Omega_X^{-1}), \qquad \Omega_X := \operatorname*{plim}_{N \to \infty} \frac{1}{N} \sum_{i_1 \ldots i_d} \eta_{i_1 \ldots i_d} \eta'_{i_1 \ldots i_d}.$$

Theorem 2 (Asymptotic distribution under heteroskedasticity)**.** *Let the assumptions in Proposition 2 hold. Additionally, let Assumptions 10-15 hold. Then, for* $N_n \to \infty$, *with* $N_n \lesssim \prod_{n' \neq n} N_{n'}$, *for all* $n = 1, \ldots, d$,

$$\sqrt{N}\big(\widehat{\beta}_{IC} - \beta^0\big) \underset{d}{\to} \mathcal{N}\left(0, \Omega_X^{-1}\, \Sigma\, \Omega_X^{-1}\right).$$

[6] See Proposition A.1 in Bai (2009) appendix

[7] See Assumption E in Section 5 in Bai (2009) for a leading example of assuming a central limit theorem in this setting.

In the presence of both heteroskedasticity and correlation in all dimensions, the most general specification to be considered here, a more general central limit theorem is required for the term, $(N)^{-1/2}\sum_{i_1\dots i_d}\eta_{i_1\dots i_d}\varepsilon_{i_1\dots i_d}$. In this case the variance is,

$$\operatorname{var}\left(\frac{1}{\sqrt{N}}\sum_{i_1\dots i_d}\eta_{i_1\dots i_d}\varepsilon_{i_1\dots i_d}\right)=\frac{1}{N}\sum_{i_1,i_1'}\cdots\sum_{i_d i_d'}\mathbb{E}(\eta_{i_1\dots i_d}\eta_{i_1'\dots i_d'})\mathbb{E}(\varepsilon_{i_1\dots i_d}\varepsilon_{i_1'\dots i_d'})$$

which includes a sequence of cross correlations for each dimension in both the η and ε terms. Following Bai (2009), this can be estimated using the Newey and West (1987) kernel approach for time dimension, in conjunction with a partial sample estimator for the cross-sectional dimensions - no theoretical analysis is provided for the consistent estimation of this variance.

Assumption 16. *(i). $\eta_{i_1\dots i_d}$ i.i.d. across $i_1\dots i_d$.*

(ii). For nonrandom positive definite $\widetilde{\Sigma}$,

$$\operatorname*{plim}_{N\to\infty}\frac{1}{N}\sum_{i_1\dots i_d}\sum_{i_1'\dots i_d'}\varepsilon_{i_1\dots i_d}\varepsilon_{i_1'\dots i_d'}\eta_{i_1\dots i_d}\eta'_{i_1'\dots i_d'}=\widetilde{\Sigma},$$

$$\frac{1}{\sqrt{N}}\sum_{i_1\dots i_d}\eta_{i_1\dots i_d}\varepsilon_{i_1\dots i_d}\xrightarrow{d}\mathcal{N}(0,\widetilde{\Sigma}).$$

The following restricts how much correlation is allowed in the error term.

Assumption 17. *Define $\sigma_{i_1\dots i_d;i_1'\dots i_d'}:=\mathbb{E}(\varepsilon_{i_1\dots i_d}\varepsilon_{i_1'\dots i_d'})$. There exists a finite $M<\infty$ such that,*

$$\xi^2_{\sigma,N}:=\frac{1}{N}\sum_{i_1'\dots i_d'\neq i_1\dots i_d}\sum_{i_1\dots i_d}\sigma^2_{i_1\dots i_d;i_1'\dots i_d'},\qquad \textit{with}\qquad \operatorname*{plim}_{N\to\infty}\xi^2_{\sigma,N}\leq M.$$

Further, for ξ_X defined in Assumption 11, $\xi_X^2\xi_{\sigma,N}N^{1/2}=o_p(1)$.

The final condition in Assumption 17 can be achieved if, e.g. $\xi_X=o_p(N^{-1/4})$, without further restrictions on $\xi_{\sigma,N}$. If $\varepsilon_{i_1\dots i_d}$ are i.i.d., or i.n.i.d., $\xi_{\sigma,N}=0$, satisfying the assumption.

Theorem 3 (Asymptotic distribution under heteroskedasticity and correlation)**.** *Let the assumptions in Proposition 2 hold. Additionally, let Assumptions 10-13, and 16-17 hold. Then, for $N_n\to\infty$, with $N_n\lesssim\prod_{n'\neq n}N_{n'}$, for all $n=1,\dots,d$,*

$$\sqrt{N}\big(\widehat{\beta}_{IC}-\beta^0\big)\underset{d}{\to}\mathcal{N}\left(0,\Omega_X^{-1}\,\widetilde{\Sigma}\,\Omega_X^{-1}\right).$$

4 Discussion of estimators

This section serves to discuss the results in Section 3, motivate further some of the chosen methods, and provide some methods to estimate proxies for kernel weights.

4.1 Matrix method results

As stated already, Proposition 1 takes for granted the dimension to flatten across admits a low rank interactive fixed-effect term for the least square method in Bai (2009). Under Assumption 1.(ii) the singular values of the flattened normalised noise term dissipates as follows:

$$\frac{1}{\sqrt{\prod_n N_n}} \left\|\varepsilon_{(n)}\right\| = O_p\left(\frac{1}{\sqrt{\min\{N_n, \prod_{m\neq n} N_m\}}}\right).$$

Interactive fixed-effect term $\mathcal{A}$ is commonly regularised in the panel data literature such that the normalised singular values are $O_p(1)$; i.e., not asymptotically negligible like those of the noise term. This ensures that, after flattening $\mathcal{A}$, each of the singular values eventually dominate those of the noise term. These conditions make up similar restrictions imposed in Ahn and Horenstein (2013) that allow for the use of the eigenvalue ratio test (ER) to diagnose the number of factors. However, this test is stipulated in a pure factor model setting, so has limited use here.

Consider the factor model applied to a flattening that may not be low-rank with the following two examples, one where the flattening is not low-rank and one where it is. First, assume $\varphi^{(1)}$ varies in a high-dimensional parameter space, e.g. with $N_1 < N_2N_3$, $\varphi^{(1)} \in \mathbb{R}^{N_1\times N_1}$ and $\Gamma \in \mathbb{R}^{N_2N_3\times N_1}$ with each column mutually orthogonal for both these matrices. Then the product of these matrices is full-rank and any factor projection approach will not fully control for this term. On the contrary, consider $\varphi^{(1)} \in \mathbb{R}^{N_1\times N_1}$ where all columns are linearly dependent. Then the matrix $\varphi^{(1)}\Gamma'$ is rank-1 regardless of L and of how $\varphi^{(2)}$ and $\varphi^{(3)}$ vary, thus can be projected with a factor model estimated with 1 factor. Hence it is important which dimension the analyst chooses to flatten over. This situation is exemplified in simulations in Section 5. Whilst requiring low-rankness in at least one dimension may be an acceptable restriction, having knowledge of which dimension this low-rankness resides in is potentially more restrictive. The kernel weighted-within transformation does not require this additional knowledge, hence is more robust.

Well established diagnostics in Bai and Ng (2002), Ahn and Horenstein (2013) and Hallin and Liška (2007) can be used to determine the number of factors in the pure factor model. These diagnostics can be repeated across different flattenings, which may be informative of the dimension to use for flattening, although no formal results are provided for using these. In the standard panel model setting, establishing the number of terms in the interactive fixed-effect in the presence of covariates is an open area of research.

Looking past the issue of knowing the multilinear rank, a standard factor model that estimates at least r_n factors results in consistent estimation of the slope coefficients, see Moon and Weidner (2015). However, for data generating processes with generic heteroskedasticity and weak correlations, the factor model can only be bounded by a slow rate of convergence. In the three dimensional setting with all dimensions roughly the same size, this bound is as slow as $N^{-1/6}$, where N is total sample size. This is too slow to use for standard inference procedures.

4.2 Estimating kernel weight proxies

Discussed here are some functionals of multidimensional arrays that are useful for estimating proxies to formulate kernel weights. This includes a discussion on how to uncover proxies from multidimensional array data, and a discussion on why standard matrix methods do not extend well to the multidimensional setting.

For the weighted-within estimator, it is important to find proxies that isolate variation in each dimension since weights are formed separately for each index. Discussed here are decompositions of multidimensional arrays that can perform this; Kolda and Bader (2009) contains a nice summary of some candidate decompositions. The method discussed here uses the higher order singular value decomposition (HOSVD), and focuses on components of this decomposition that have well formulated theoretical properties. The HOSVD is traditionally used in pursuit of a low-rank tensor decomposition by either direct truncation of left singular vectors or by some iteration approach similar to this; see for example the higher order orthogonal iteration scheme. The problem of direct truncation, however, is not well-posed because the solution to the low-rank tensor problem may not be unique and reformulating the original tensor after the aforementioned truncation is not guaranteed to be lower tensor rank. See De Silva and Lim (2008) for an extensive explanation of the ill-posedness issues. Hence, this method cannot be used in the pursuit of analytic consistency results. Problems also arise in this setting where the reformulated tensors can be arbitrarily well approximated by a tensor of lower tensor rank, which is a result of the border rank issue of the tensor rank decomposition.

Reconsider the three dimensional model with heterogeneity of the following form

$$\mathcal{A}_{ijt} = \sum_{\ell=1}^{L} \varphi_{i\ell}^{(1)} \varphi_{j\ell}^{(2)} \varphi_{t\ell}^{(3)}. \tag{15}$$

As shown in De Silva and Lim (2008), the Eckart–Young–Mirsky theorem cannot be relied upon to guarantee an optimal low-rank approximation for the multidimensional array $\boldsymbol{\mathcal{A}}$. Reconsider the SVD for matrices, applied to each of the n-flattenings of $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{N_1 \times, \ldots, \times N_d}$ as

$$\mathcal{A}_{(n)} = U^{(n)} \Sigma_n V^{(n)\prime}. \tag{16}$$

By the same logic as in the matrix case and formalised with the Eckart-Young-Mirsky theorem, variation over the rows of each $U^{(n)}$ explains variation over the n^{th} dimension of the multidimensional array $\boldsymbol{\mathcal{A}}$. Thus, if $\mathcal{A}_{(n)}$ is low rank, the leading few columns of $U^{(n)}$ provide good proxies for closeness in n^{th} dimension of $\boldsymbol{\mathcal{A}}$. If $\mathcal{A}_{(n)}$ is not low rank then these leading columns still provide good proxies for bias reduction using the kernel-weighted fixed-effects. Hence, by reconsidering the tensor problem as a sequence of matrix problems, the usual singular value decomposition properties can be utilised for this bias reduction.

Consider for any of the dimensions n the corresponding matrix of left singular vectors from above, $U^{(n)}$, estimated with noise ε_{ijt}. That is, each $U^{(n)}$ are calculated from the matrix $\boldsymbol{\mathcal{V}}_{(n)} = \boldsymbol{\mathcal{A}}_{(n)} + \boldsymbol{\varepsilon}_{(n)}$. Under regularity conditions on the noise term $\boldsymbol{\varepsilon}$, the left singular vectors from this decomposition consistently estimate $U^{(n)}$, up to scalings and rotations. For example, in the three dimensional case, define the r_1-vector $\widehat{U}_i^{(1)}$ as the i-th row of the left singular matrix of $\boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon}$ flattened in the first dimension. Then the vector $\widehat{U}_i^{(1)}$ may comprise of,

$$\widehat{U}_i^{(1)} = \varphi_i^{(1)} + O_p\left(\frac{1}{\sqrt{\min\{N_1, N_2N_3\}}}\right).$$

Likewise, for any dimension n define $\widehat{U}^{(n)}$ as the matrix of singular vectors from $\boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon}$ flattened in the n-th dimension, where $\boldsymbol{\mathcal{A}}$ is the unobserved fixed-effects component of interest and $\boldsymbol{\varepsilon}$ is the usual idiosyncratic noise term. Then, Bai and Ng (2002) detail conditions required for the following "up-to-rotation" consistency result, which has been amended to this paper's setting;

Lemma 1 (Theorem 1 from Bai and Ng (2002))**.** *For any fixed integer $k \geq 1$, there exists an $(r_n \times k)$ matrix H_n^k with $\operatorname{rank}(H_n^k) = \min\{k, r_n\}$ and $C_n = \min\left\{\sqrt{N_n}, \prod_{n' \neq n} \sqrt{N_{n'}}\right\}$ such that for each n under some regularity conditions*

$$C_n^2 \left\| \widehat{U}_{i_n}^{(n)} - H_n^{k\prime} \varphi_{i_n}^{(n)} \right\|^2 = O_p(1).$$

This establishes a consistency result for estimating weight proxies and suggests these left singular vectors are viable options to input into kernels to formulate weights in each dimension if the true error, $\boldsymbol{\mathcal{V}} = \boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon}$, is observed. It also makes concrete the limitation implied by the value of C_n for each index - that short indices have poorly estimated proxies. Given that the fixed-effect approximation error term for the weighted-within transformation is multiplicative across dimension, the error from this relatively poor approximation becomes negligible as long as enough other dimension proxies are well estimated. Also, the presence of the rotation matrices, H_n^k, in Lemma 1 can be ignored since these do not change relative distances of each unit under standard distance metrics used in the kernel weights.

However, it is not necessarily justified to assume the true error, $\boldsymbol{\mathcal{V}} = \boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon}$, is observed. In fact, an estimate of the error $\widehat{\mathcal{V}}_{ijt} = Y_{ijt} - X_{ijt}\widehat{\beta} = X_{ijt}(\beta^0 - \widehat{\beta}) + \mathcal{A}_{ijt} + \varepsilon_{ijt}$, clearly depends on the estimate $\widehat{\beta}$, hence should be bound by the rate of convergence for this estimate. For this reason, the uniform convergence result from Lemma 1 is unlikely to be useful. Proposition A.1

in Bai (2009) does provides for the mean squared deviation,

$$\frac{1}{N_n}\sum_{i_n=1}^{N_n}\left\|\widehat{U}_{i_n}^{(n)} - H_n^{k\prime}\varphi_{i_n}^{(n)}\right\|^2 = O_p(\|\beta^0 - \widehat{\beta}\|^2) + O_p\left(\frac{1}{\min\{N_n, \prod_{n'\neq n} N_{n'}\}}\right),$$

if $\widehat{U}_{i_n}^{(n)}$ come from least squares in Bai (2009). Ignoring H_n^k, $\frac{1}{N_n}\sum_{i_n=1}^{N_n}\left\|\widehat{U}_{i_n}^{(n)} - \varphi_{i_n}^{(n)}\right\|^2 = O_p\left({N_n}^{-1}\right)$ if each dimension sample size grows at a roughly comparable rate and the convergence rate for $\|\beta^0 - \widehat{\beta}\|$ in Proposition 1 is used.[8] Then, C_n^{-2} from Proposition 2 is $1/N_n$ and the parametric rate for the kernel weighted estimator can be established.

The invertible matrix H_n^k can be ignored in the above display because these preliminary estimates are used to form distance measure between units. Hence, since the true distance of interest is that between the true parameter values,

$$\left\|\varphi_{j_n}^{(n)} - \varphi_{i_n}^{(n)}\right\|^2 = \left\|(H_n^{k\prime})^{-1}\left(H_n^{k\prime}\varphi_{j_n}^{(n)} - H_n^{k\prime}\varphi_{i_n}^{(n)}\right)\right\|^2 \leq \frac{1}{\lambda_{\min}(H_n^{k\prime})}\left\|H_n^{k\prime}\varphi_{j_n}^{(n)} - H_n^{k\prime}\varphi_{i_n}^{(n)}\right\|^2$$

where the inequality is a spectral norm bound, and $\lambda_{\min}(H_n^{k\prime})$ is the smallest eigenvalue of $H_n^{k\prime}$, which is strictly positive from invertibility. Therefore, it is without loss of generality to bound the space of true parameters, or the space of true parameters once rotated by $H_n^{k\prime}$.

Notice, for the space of the proxy measures - $\widehat{U}_{i_n}^{(n)}$ in the above example - to span the space of the true fixed-effects $\varphi_{i_n}^{(n)}$, it is necessary that the product space of fixed-effects over $n' \neq n$ has rank greater than $\varphi^{(n)}$. This is not, however, problematic in the context of this paper. Take the example where the $\widehat{U}_{i_n}^{(n)}$ indeed spans a lower rank space than the ambient space of $\varphi_{i_n}^{(n)}$, but are still estimates of linear combinations of $\varphi_{i_n}^{(n)}$. This could happen in the situation where, for example, the space of $\varphi^{(n)}$ has dimension greater than the product space over $\varphi^{(n')}$ for $n' \neq n$. Then $\widehat{U}_{i_n}^{(n)}$ actually estimates $U_{i_n}^{(n)} = \widetilde{H}\varphi_{i_n}^{(n)}$, a linear combination where $\text{rank}(\widetilde{H}) = r_n < \text{rank}(\varphi^{(n)})$. The reason this is not problematic here is that the normed difference $\|\varphi_{i_n}^{(n)} - \varphi_{i'_n}^{(n)}\|$ can be bounded by $\|\widehat{U}_{i_n}^{(n)} - \widehat{U}_{i'_n}^{(n)}\|$ and some additional fixed-effect estimation errors. Hence, the results still carry through for the case where only lower-dimensional linear combinations of fixed-effects can be identified and estimated, see Lemma A.2.

5 Simulations

Two simulation exercises are performed. The first analyses the performance of the matrix methods compared to the kernel weighted methods when sample size is allowed to increase. These results are summarised in Figure 1 and Table 1. The second simulation exercise analyses

[8] To be precise, the comparable rate restriction for any dimension $n = 1, \ldots, d$ is $N_n \lesssim \prod_{n'\neq n} N_{n'}$. The comparable rate commonly used in the two-dimensional panel literature is $N \approx T$. Hence, in the multidimensional problem, the relative rate growth is milder than the two-dimensional case.

some fixed sample properties of the set of estimators when Assumption 3 is violated in some dimensions. This assumption governs the rank of the fixed-effect term after it is transformed into a matrix. These results are summarised in Table 2.

5.1 Growing sample exercise

Data is generated as,

$$
\begin{aligned}
Y_{ijt} &= X_{ijt} + \mathcal{A}_{ijt} + \varepsilon_{ijt} \\
X_{ijt} &= 2\mathcal{A}_{ijt} - \rho \sum_{\ell=1}^{2} \Big(\lambda_{i\ell} + \lambda_{i-1\ell}\Big)\Big(\gamma_{j\ell} + \gamma_{j-1\ell}\Big)\Big(f_{t\ell} + f_{t-1,\ell}\Big) + \eta_{ijt} \\
\varepsilon_{ijt} &= \Big(1/\sqrt{2}\Big)\big(\nu_{ijt} + \nu_{ijt-1} + \nu_{ij-1t} + \nu_{ij-1t-1} + \nu_{i-1jt} + \nu_{i-1jt-1} + \nu_{i-1j-1t} + \nu_{i-1j-1t-1}\big)
\end{aligned} \tag{17}
$$

with, $\mathcal{A}_{ijt} = \sum_{\ell=1}^{2} \lambda_{i\ell}\gamma_{j\ell}f_{t\ell}$. Also,

$$\eta_{ijt} \overset{i.i.d.}{\sim} N(0,1), \nu_{ijt} \overset{i.ni.d.}{\sim} N(0, \min\{4, \eta_{ijt}^2\}) \text{ and for each } \ell,\ \lambda_{i\ell}, \gamma_{j\ell}, f_{t\ell} \overset{i.i.d.}{\sim} N(0,1).$$

ε_{ijt} admits heteroskedasticity, and correlation within each dimension of the data. The order of each i, respectively j, is randomised to simulate an unknown cross correlation structure.

The results depicted in Figure 1 show two specifications for ρ. The left panel depicts $\rho = 1$, and the right panel depicts $\rho = 1/3$. Estimators considered use 2 estimated factors such that the multilinear rank is correctly predicted. The left panel shows that whilst the matrix methods, labelled Factor, look consistent, they converge at a rate much slower than the empirical bounds. This leads to spurious inference. The right panel shows more egregious convergence when the bias problem is made worse with $\rho = 1/3$. Indeed, for the sample sizes considered, the empirical bounds do not ever cover the true parameter value in either panel.

This contrasts with the results for the kernel weighted fixed-effect projection. Finite sample bias is negligible compared to variance in both the left and right panel. This estimator has roughly the same variance as the matrix methods, but for the sample sizes considered, the empirical bounds are always valid.

Table 1 displays coverage and compares the kernel weighted estimator without and with the inference correction from Section 3.3, and the factor model in any dimension. The Ker Inf coverage, which is coverage for the inference corrected estimator, is always better than the uncorrected estimator. This follows directly from the theoretical arguments made in Section 3.3. The factor model undercovers with progressively worse coverage as the sample size grows.

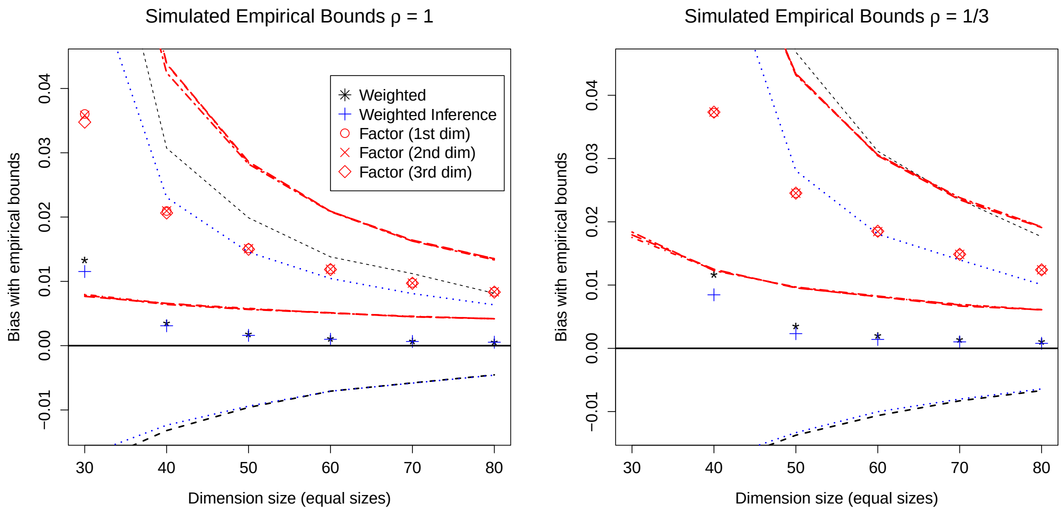

Figure 1: Bias of β estimates with 95% empirical bounds for DGP in (17)

Left panel, $\rho = 1$, right panel $\rho = 1/3$. 10,000 Monte Carlo rounds. Points depict mean bias. Dotted lines depict 95% empirical bounds. Factor # depicts the matrix factor methods with dimension # as rows. Weighted and weighted inference depicts the kernel weighted method, and kernel weighted method with inference correction.

5.2 Fixed sample exercise

Table 2 shows simulation results for the following DGP,

$$Y_{ijt} = X_{ijt}\beta + \mathcal{A}_{ijt} + \varepsilon_{ijt}$$

$$X_{ijt} = \mathcal{A}_{ijt} + \sum_{\ell=1}^{N_1} \Big(\lambda_{i\ell} + \lambda_{i-1\ell}\Big)\Big(\gamma_{j\ell} + \gamma_{j-1\ell}\Big)\Big(f_{t\ell} + f_{t-1,\ell}\Big) + \eta_{ijt}$$

with $\mathcal{A}_{ijt} = \sum_{\ell=1}^{N_1} \lambda_{i\ell}\gamma_{j\ell}f_{t\ell}$ and all other parameters the same as (17). $\mathcal{A}_{ijt}$ is normalised to have unit variance. $\boldsymbol{\mathcal{A}}$ is specified such that it is rank 1 when flattened in the first dimension and rank N_1 when flattened in either dimension two or three. That is, the multilinear rank is $\mathbf{r} = (1, N_1, N_1)$. This comes directly from the data generating process for each $\varphi^{(n)}$, where the matrix $\varphi^{(1)}$ is designed to be rank-1 and the matrices $\varphi^{(2)}$ and $\varphi^{(3)}$ are designed to be rank-N_1.

In Table 2, the estimators OLS and Fixed-effects are simply the pooled OLS estimator and the pooled OLS estimator after additive fixed-effects are projected out, respectively. As expected both of these have bias. The factor model is used after first flattening along each dimension as Factor(dim $= n$), where n is the dimension used for flattening. In each case, 2 factors are projected. The results show the bias is close to zero when the correct dimension is flattened over (the first dimension in this case) and very poor bias when the incorrect dimension is used (the second and third dimensions). Lastly, the kernel differencing estimator is estimated with Gaussian kernel function with various bandwidths; which are standardised to be equivalent to

	$\widehat{r} = 2$		Coverage		
Dim	Ker	Ker Inf	Factor 1	Factor 2	Factor 3
10	0.04	0.19	0.11	0.11	0.17
20	0.20	0.50	0.07	0.07	0.14
30	0.48	0.85	0.04	0.05	0.12
40	0.56	0.92	0.03	0.03	0.09
50	0.57	0.93	0.02	0.02	0.08
60	0.58	0.95	0.01	0.01	0.06
70	0.58	0.95	0.00	0.00	0.05
80	0.60	0.95	0.00	0.00	0.04

Table 1: Estimated coverage for %5 nominal; test 10,000 Monte Carlo rounds
Ker stands for the kernel weighted estimator without inference correction. Ker Inf stands for the kernel weighted estimator with the inference correction. Factor # is the matrix method making dimension # the rows, and the other dimensions jointly the columns. Dim stands for the size of each dimension, such that the total sample size is the cube of this. Coverage is measured for heteroskedasticity and correlation robust standard errors.

standard deviations of the proxy measures. Kernel estimators show a clear bias-variance trade-off, but all with bias of the same order as the correct factor model.

This analysis is repeated for the four dimensional case in Table 2, where the first and second dimensions admit low-dimensional unobserved interactive fixed-effects parameters. The simulations suggest similar results as the three dimensional case, where the factor models perform well when flattened in the low-dimensional dimensions (first and second) and poorly in the high-dimensional dimensions (third and fourth).

3-D	Bias	St. dev.	RMSE	4-D	Bias	St. dev.	RMSE
OLS	0.3655	0.0038	0.3655		0.1190	0.0211	0.1209
Fixed-effects	0.3708	0.0040	0.3708		0.1321	0.0265	0.1347
Kernel Inf (h = 0.05)	0.0017	0.0083	0.0085		-0.0002	0.0068	0.0068
Kernel Inf (h = 0.1)	0.0035	0.0060	0.0069		0.0003	0.0040	0.0040
Kernel Inf (h = 0.2)	0.0090	0.0050	0.0103		0.0027	0.0031	0.0041
Factor (dim = 1)	-0.0028	0.0041	0.0050		0.0013	0.0016	0.0021
Factor (dim = 2)	0.3603	0.0064	0.3604		-0.0013	0.0016	0.0021
Factor (dim = 3)	0.3603	0.0064	0.3604		0.1199	0.0256	0.1226
Factor (dim = 4)	NA	NA	NA		0.1205	0.0269	0.1234

Table 2: Fixed-sample size simulation with heterogeneous multilinear rank
3D model ($N_1 = N_2 = N_3 = 40$), with 10,000 Monte Carlo rounds.
4D model ($N_1 = N_2 = N_3 = N_4 = 20$), with 1,000 Monte Carlo rounds. Bandwidth scaled up by $\sqrt{2}$ to account for smaller per dimension sample. All results relate to β estimation.

6 Empirical application - demand estimation for beer

The methods proposed in this paper are applied to estimate the demand elasticity for beer. Price and quantity for beer sales is taken from the Dominick's supermarket dataset for the years 1991-1995 and is related to supermarkets across the Chicago area. Price and quantity vary over three dimensions in this example – product (i), store (j) and fortnights (t). Fixed-effects that interact across all three dimensions can control for temporal taste shocks to beer consumption that differ over both product and store. Take for instance a large sporting event (temporary t shock) that changes preferences differently across locations (j) and across certain subsets of sponsored beer (i). Analysts may want some way to control for such heterogeneity that interacts over all dimensions, even if they do not explicitly observe such sponsorship variables. For example, in the stadiums for the many NBA finals playoffs the Chicago Bulls played in the early 1990's, Miller Lite beer advertisements could be seen alongside advertisements for a substitute product, Canadian Club whisky. This suggests these events attracted large marketing campaign spends for these and other beer substitute brands that most likely also included price offers at local supermarkets. Whilst the impact of these advertisements and price offers on the demand for or price of beer is not clear and, further, that it is reasonably safe to assume the econometrician does not observe the plethora of marketing campaigns around these events, the analyst would most likely still want to control for aggregate shocks like these.

Models for demand estimation ideally account for endogenous variation in prices and quantity. The classic instrumental variable approach is to find a supply shifter that shifts the supply curve, allowing the econometrician to trace out the slope of the demand curve. A popular instrument in the estimation of beer demand is the commodity price for barley, one of the product's main ingredients, see e.g. Saleh (2014); Tremblay and Tremblay (1995); Richards and Rickard (2021). Since the price of barley is arguably not driven by the demand for it by any one supplier of beer, it can be a useful variable to instrument for price shifts. In the following, it is taken as given that the price of barley is exogenous with respect to the fixed-effects and noise term, ε.

For valid inference, the instrument is also required to be strong, in the sense that it is strongly correlated with the price of beer. In this dataset, correlation between the price of barley, which varies over only month, every second t, and price of beer depends on how beer price is first aggregated. If beer price is first integrated over i, j, and to the monthly level, such that it only varies over every second t, then it is highly correlated with the price of barley, at 0.79. However, if beer price is not aggregated at all it is only correlated at 0.001. This heuristic suggests there are important product and store level price drivers for beer that are not accounted for by fluctuations in the price of barley, or indeed by price fluctuations only over time. The price of barley is only observed at the monthly level, which further reduces variability after projecting beer prices onto this instrument. Standard errors for the IV estimator below are much larger than other estimators, which may be explained partly by this loss in effective sample size.[9]

Table 3 refers to the estimates for demand elasticities for the following regression model,

$$\log(quantity_{ijt}) = \log(price_{ijt})\beta + \mathcal{A}_{ijt} + \varepsilon_{ijt} \tag{18}$$

where $\mathcal{A}_{ijt}$ is the usual interactive fixed-effects term from the prequel. This amounts to estimating the standard log-log model for demand with fixed-effects. That is,

$$quantity_{ijt} = price_{ijt}^{\beta} \exp(\mathcal{A}_{ijt} + \varepsilon_{ijt}).$$

No additional controls are included here since they are low-dimensional and subsumed by the fixed-effects term. Estimates for pooled OLS are positive, a contradiction that demand curves are downward sloping. Whilst IV estimates are negative, standard errors are very large. As noted above, the effective sample size for the second stage of the instrumental variable approach is orders of magnitude smaller because the instrument only varies over time, specifically every second fortnight. Estimates under the additive model for fixed-effects are also negative, with much smaller standard errors than the IV estimate.

The matrix methods, labelled Factor, are implemented in all three dimensions of the data. That is, the three-dimensional array is transformed into three different matrices, one with products as rows, one with stores as rows, and lastly with time as rows. In all three specifications,

[9]The effective sample size for the IV estimator drops from $N_1 N_2 T$ to just T.

five factors are estimated. The results suggest the factor model for fixed-effects can be sensitive to how the array is organised into a two-dimension problem. Simulation results from Section 5 suggest a potential explanation - sparsity of the dimension specific fixed-effects in the interaction term can be heterogeneous, and imply a different factor model rank when the tensor of data is flattened, or matricised, in different dimensions. This in turn implies completely different models when the data is transformed these different ways. The kernel weighted fixed-effects estimates elasticities similar to the IV point estimates, but with much smaller standard errors.

Estimates of the same log-log model controlling for average log price of other producer products are numerically very similar, hence omitted. The kernel weighted estimates are similar to the own-price elasticity estimates from Table 1 in Hausman, Leonard and Zona (1994), which average around -2.5.

Figure 2 displays estimates from the factor model with products as rows, and the weighted-within estimator, with estimated confidence bands, allowing the estimated number of interactive fixed-effects terms to increase. The figure displays two things. First, there is evidence that up to approximately 9-10 interactive terms may be appropriate. Second, that at each number of estimated interactive terms, there is a substantial shift in estimate when the weighted-within transformation is applied. Taking the model and theory proposed in this paper for granted, this implies a persistent debias with the weighted-within transformation.

Estimator	$\widehat{\beta}$ (St. err.)
Pooled OLS	1.304 (0.007)
Additive Fixed-effects	-0.069 (0.040)
Pooled IV	-2.148 (1.158)
Factor (Product (i) as rows)	-2.787 (0.055)
Factor (Store (j) as rows)	-0.098 (0.042)
Factor (Time (t) as rows)	-0.033 (0.003)
Weighted-within	-3.317 (0.053)

Table 3: Log-log demand elasticities (45 products, 48 stores, 110 fortnights).
Estimates for the standard errors are included in brackets. The factor model estimator uses five factors in all three cases.

7 Conclusion

This paper develops methods to generalise the interactive fixed-effect to multidimensional datasets with more than two dimensions. Theoretical results show that standard matrix methods can be

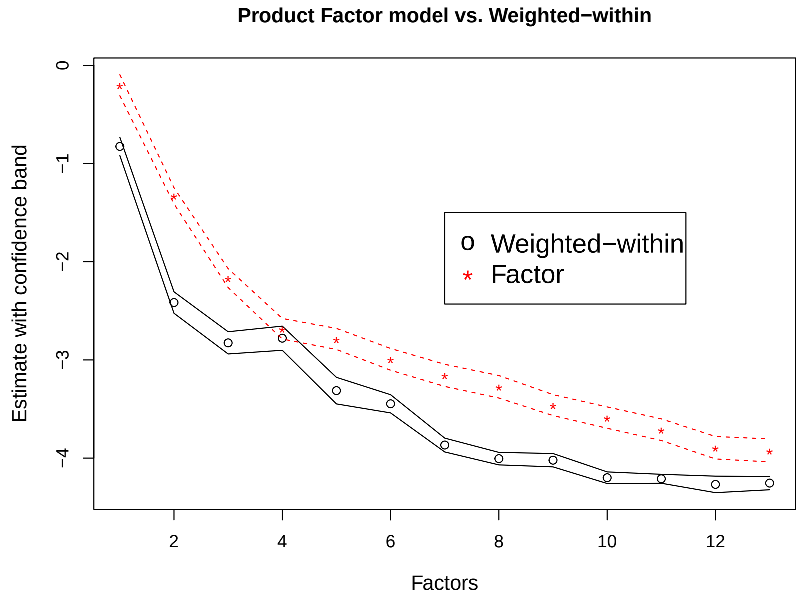

Figure 2: Elasticity estimates for increasing number of estimated factors

applied to this setting but require additional knowledge of the data generating process and generally have slow convergence rates. Nonetheless, these provide useful preliminary estimates. The multiplicative interactive fixed-effect error from the kernel weighted method show an improvement on the convergence rate of slope coefficient estimates to the parametric rate, and suggest a more robust approach to projecting fixed-effects. Simulations show finite sample properties when the sample size is allowed to grow and when it is fixed. These simulations exemplify how sensitive the two-dimensional methods are to model specification issues as simple as how to organise the dataset. They also show the robustness of the kernel weighted fixed-effects estimators without having to make these same specifications. A method for inference with the weighted fixed-effects estimator is also introduced.

The model is applied to a demand model for beer consumption. The application demonstrates that simply applying the two-dimensional factor model approach is sensitive to how dimensions are arranged to suit these estimators. The weighted-within transformation estimates elasticities close to an instrumental variable point estimate, but with substantially better precision.

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A Appendix: Proofs

A.1 Proofs of preliminary results

Lemma A.1 (Regularity of proxy measures)**.** *Let $\widehat{\varphi}_{i_n}^{(n)} \in \widehat{\Phi}_n$ be the proxy space for the fixed-effects and let $k(\cdot)$ be a bounded kernel function from Assumption 5. If Assumption 6 holds, then as $h_n \to 0$ such that $N_n h_n \to \infty$,*

$$\underset{N_n\to\infty}{\text{plim}} \frac{1}{N_n} \sum_{i_n=1}^{N_n} \sum_{j_n \neq i_n} k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\|\right) > 0 \qquad wpa1. \tag{A.1}$$

This implies

$$N_n^{-1} Tr\{(\mathbb{I}_{N_n} - W_n)'(\mathbb{I}_{N_n} - W_n)\} > 0 \qquad wpa1. \tag{A.2}$$

Proof of Lemma A.1. The supposition is that when

$$\operatorname*{plim}_{N_n\to\infty} \frac{1}{N_n}\sum_{i_n=1}^{N_n} M_n^h\left(\widehat{\varphi}_{i_n}^{(n^*)}\right) > 0 \qquad \text{wpa1.} \tag{A.3}$$

in Assumption 6 holds, then (A.1) holds. Take

$$\begin{aligned}
\frac{1}{N_n}\sum_{i_n=1}^{N_n}\sum_{j_n\neq i_n} k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)}-\widehat{\varphi}_{j_n}^{(n)}\right\|\right) &= \frac{1}{N_n}\sum_{i_n=1}^{N_n}\sum_{j_n:\widehat{\varphi}_{j_n}^{(n)}\in B_{h\varepsilon}(\widehat{\varphi}_{i_n}^{(n)})} k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)}-\widehat{\varphi}_{j_n}^{(n)}\right\|\right)\\
&+\frac{1}{N_n}\sum_{i_n=1}^{N_n}\sum_{j_n:\widehat{\varphi}_{j_n}^{(n)}\notin B_{h\varepsilon}(\widehat{\varphi}_{i_n}^{(n)})} k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)}-\widehat{\varphi}_{j_n}^{(n)}\right\|\right)\\
&=: A_1 + A_2
\end{aligned}$$

If the probability limit of A_1 or A_2 is strictly positive, then the sum is bounded below since both terms are weakly positive.

$$A_1 \geq \frac{1}{N_n}\sum_{i_n=1}^{N_n}\sum_{j_n\neq i_n} \mathbb{1}\{\widehat{\varphi}_{i_n}^{(n)}\in B_{h\varepsilon}(\widehat{\varphi}_{j_n}^{(n)})\}k\left(\varepsilon\right).$$

Set ε low enough such that $k(\varepsilon) > 0$. Then $\operatorname{plim} A_1 > 0$ by (A.3). $A_2 \geq 0$, hence no tighter lower bound needs to be shown for this term.

Since weights are in $[0,1]$ and sum to 1 only the trace of $N_n^{-1}(\mathbb{I}_{N_n}-W_n)^2$ needs to be bounded away from 0 for (A.2) to hold. Using shorthand $K_{ij,h}^{(n)} = k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)}-\widehat{\varphi}_{j_n}^{(n)}\right\|\right)$,

$$\frac{1}{N_n}Tr\{(\mathbb{I}_{N_n}-W_n)^2\} = \frac{1}{N_n}\sum_{i_n}(1-W_{n,i_ni_n})^2 > 0 \qquad \text{wpa1.}$$

if it can be shown that $\frac{1}{N_n}\sum_{i_n}\sum_{j_n\neq i_n} W_{n,j_ni_n} > 0$ wpa1. The first part of the Lemma showed $\frac{1}{N_n}\sum_{i_n}\sum_{j_n\neq i_n} K_{ij,h}^{(n)} > 0$ wpa1, which shows (A.2) holds.

■

Lemma A.2 (Linear Combinations Bound)**.** *For the linear operator $\widetilde{H}_n\in\mathbb{R}^{r_n\times L}$, with bounded eigenvalues, define linear transformations of $\varphi_{i_n}^{(n)}$ as $\widetilde{\varphi}_{i_n}^{(n)} = \widetilde{H}_n\varphi_{i_n}^{(n)}$. When rank$(\widetilde{H}_n) = r_n < L$,*

$$\|\varphi_{i_n}^{(n)}-\varphi_{i_n'}^{(n)}\|^2 \leq \lambda_{\max}(\widetilde{H}_n)\left(\|\widetilde{\varphi}_{i_n}^{(n)}-\widehat{\widetilde{\varphi}}_{i_n}^{(n)}\|^2+\|\widetilde{\varphi}_{i_n'}^{(n)}-\widehat{\widetilde{\varphi}}_{i_n'}^{(n)}\|^2+\|\widehat{\widetilde{\varphi}}_{i_n}^{(n)}-\widehat{\widetilde{\varphi}}_{i_n'}^{(n)}\|^2\right)$$

where $\lambda_{\max}$ is the maximum eigenvalue. When rank$(\widetilde{H}_n) = r_n = L$ then $\widetilde{H}_n$ is invertible and bounding $\|\varphi_{i_n}^{(n)}-\varphi_{i_n'}^{(n)}\|^2$ by $\|\widetilde{\varphi}_{i_n}^{(n)}-\widetilde{\varphi}_{i_n'}^{(n)}\|^2$ is straightforward.

Proof of Lemma A.2.

$$\begin{aligned}\|\varphi_{i_n}^{(n)} - \varphi_{i_n'}^{(n)}\|^2 \le\ & 5\|\varphi_{i_n}^{(n)} - \widetilde{H}_n'\widetilde{\varphi}_{i_n}^{(n)}\|^2 + 5\|\varphi_{i_n'}^{(n)} - \widetilde{H}_n'\widetilde{\varphi}_{i_n'}^{(n)}\|^2 \\ &+ 5\|\widetilde{H}_n'\widetilde{\varphi}_{i_n}^{(n)} - \widetilde{H}_n'\widehat{\widetilde{\varphi}}_{i_n}^{(n)}\|^2 + 5\|\widetilde{H}_n'\widetilde{\varphi}_{i_n'}^{(n)} - \widetilde{H}_n'\widehat{\widetilde{\varphi}}_{i_n'}^{(n)}\|^2 \\ &+ 5\|\widetilde{H}_n'\widehat{\widetilde{\varphi}}_{i_n}^{(n)} - \widetilde{H}_n'\widehat{\widetilde{\varphi}}_{i_n'}^{(n)}\|^2\end{aligned}$$

The first term from the right hand side, $\|\varphi_{i_n}^{(n)} - \widetilde{H}_n'\widetilde{\varphi}_{i_n}^{(n)}\|^2 = \|(\mathbb{I}_L - \widetilde{H}_n'\widetilde{H}_n)(\varphi_{i_n}^{(n)} - \varphi_{i_n}^{(n)})\|^2 = 0$, from $\|(\mathbb{I}_L - \widetilde{H}_n'\widetilde{H}_n)(\varphi_{i_n}^{(n)} - \varphi_{i_n}^{(n)})\|^2 \le \lambda_{\max}(\mathbb{I}_L - \widetilde{H}_n'\widetilde{H}_n)\|\varphi_{i_n}^{(n)} - \varphi_{i_n}^{(n)}\|^2 = 1 \cdot \|\varphi_{i_n}^{(n)} - \varphi_{i_n}^{(n)}\|^2$. Likewise, the second term is 0. Finally, use the matrix norm bound to bound the three remaining terms by the maximum eigenvalue of $\widetilde{H}_n$ factored out from each term. ■

The remaining proofs implicitly use Lemma A.2 and bound the space of each $\varphi_{i_n}^{(n)}$ without loss of generality with regard to the linear combination space that may be identified and estimated in the matrix method.

A.2 Proofs of main paper results

Proof of Proposition 2. In the following, let $\text{vec}_K(\widetilde{\mathbf{X}})$ be the $\prod_n N_n \times K$ matrix of vectorised covariates after the within-cluster transformations where each column is a vectorised transformed covariate. The vec operator on other variables is the standard vectorisation operator. Also let $N = \prod_n N_n$ and the subscript $i = 1, \ldots, N$ be the index for the vectorised data when i has no subscript. Then,

$$\begin{aligned}\beta_{KFE,\mathcal{C}} &= \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}(\widetilde{\mathbf{Y}}) \\ &= \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})' \left(\text{vec}_K(\widetilde{\mathbf{X}})\beta^0 + \text{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) + \text{vec}(\widetilde{\boldsymbol{\varepsilon}})\right) \\ &= \beta^0 + \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})' \left(\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) + \text{vec}(\widetilde{\boldsymbol{\varepsilon}})\right),\end{aligned}$$

such that,

$$\begin{aligned}\left\|\beta_{KFE,\mathcal{C}} - \beta^0\right\| &= \left\| \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})' \left(\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) + \text{vec}(\widetilde{\boldsymbol{\varepsilon}})\right) \right\| \\ &\le \|\kappa_N\| + \|\omega_N\|\end{aligned}$$

where

$$\begin{aligned}\|\kappa_N\| &:= \left\| \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}})\right\|; \\ \|\omega_N\| &:= \left\| \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}(\widetilde{\boldsymbol{\varepsilon}})\right\|.\end{aligned}$$

$\|\omega_N\|$ is bounded at the parametric rate by standard argumetns, so $\|\kappa_N\|$ is the focus here.

Notice,

$$\|\kappa_N\| \leq \left\| \left(\mathrm{vec}_K(\widetilde{\mathbf{X}})'\mathrm{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \right\| \left\|\mathrm{vec}_K(\widetilde{\mathbf{X}})'\mathrm{vec}(\widetilde{\boldsymbol{\mathcal{A}}})\right\|$$

Focus on the right hand part, and let $\langle .,. \rangle_F$ be the Frobenius inner product,

$$\left\|\mathrm{vec}_K(\widetilde{\mathbf{X}})'\mathrm{vec}(\widetilde{\boldsymbol{\mathcal{A}}})\right\| = \left\| \begin{bmatrix} \langle \widetilde{X}_1, \widetilde{\boldsymbol{\mathcal{A}}} \rangle_F \\ \vdots \\ \langle \widetilde{X}_K, \widetilde{\boldsymbol{\mathcal{A}}} \rangle_F \end{bmatrix} \right\| \leq \left\| \begin{bmatrix} \left|\sum_{i=1}^N \widetilde{X}_{i,1}\widetilde{\mathcal{A}}_i\right| \\ \vdots \\ \left|\sum_{i=1}^N \widetilde{X}_{i,K}\widetilde{\mathcal{A}}_i\right| \end{bmatrix} \right\| \tag{A.4}$$

Let $\bar{\varphi}^{(n)}_{i_n^*,\ell}$ denote now the estimate of $\varphi^{(n)}_{i_n,\ell}$ according the the kernel weighted estimator. The entries for each k in (A.4) can be written as,

$$\begin{aligned}
\sum_{i=1}^N \widetilde{X}_{i,k}\widetilde{\mathcal{A}}_i &= \frac{1}{N}\sum_{i_1,\dots,i_d} X_{i_1,\dots,i_d;k} \sum_{\ell=1}^L \prod_n \left(\varphi^{(n)}_{i_n,\ell} - \bar{\varphi}^{(n)}_{i_n^*,\ell}\right) \\
&= \frac{1}{N}\sum_{i_1,\dots,i_d} X_{i_1,\dots,i_d;k} \sum_{\ell=1}^L \prod_n \sum_{j_n} W^{(n)}_{i_n,j_n} \left(\varphi^{(n)}_{i_n,\ell} - \varphi^{(n)}_{j_n,\ell}\right) \\
&= \frac{1}{N}\sum_{i_1,\dots,i_d}\sum_{j_1,\dots,j_d}\sum_{\ell=1}^L \prod_n \sqrt{W^{(n)}_{i_n,j_n}} X_{i_1,\dots,i_d;k} \prod_n \sqrt{W^{(n)}_{i_n,j_n}} \left(\varphi^{(n)}_{i_n,\ell} - \varphi^{(n)}_{j_n,\ell}\right) \\
&\leq \frac{1}{N}\left(\sum_{i_1,\dots,i_d}\sum_{j_1,\dots,j_d}\sum_{\ell=1}^L \prod_n W^{(n)}_{i_n,j_n} X^2_{i_1,\dots,i_d;k}\right)^{1/2} \times \dots \\
&\dots \times \left(\sum_{i_1,\dots,i_d}\sum_{j_1,\dots,j_d}\sum_{\ell=1}^L \prod_n W^{(n)}_{i_n,j_n} \left(\varphi^{(n)}_{i_n,\ell} - \varphi^{(n)}_{j_n,\ell}\right)^2\right)^{1/2} \\
&\leq \left(\frac{1}{N}\sum_{\ell=1}^L \sum_{i_1,\dots,i_d} X^2_{i_1,\dots,i_d;k} \sum_{j_1,\dots,j_d} \prod_n W^{(n)}_{i_n,j_n}\right)^{1/2} \times \dots \\
&\dots \times \left(\frac{1}{N}\sum_{i_1,\dots,i_d}\sum_{j_1,\dots,j_d} \prod_n W^{(n)}_{i_n,j_n} \left\|\varphi^{(n)}_{i_n} - \varphi^{(n)}_{j_n}\right\|^2\right)^{1/2} \\
&= \sqrt{L}O_p(1)\left(\frac{1}{N}\sum_{i_1,\dots,i_d}\sum_{j_1,\dots,j_d} \prod_n W^{(n)}_{i_n,j_n} \left\|\varphi^{(n)}_{i_n} - \varphi^{(n)}_{j_n}\right\|^2\right)^{1/2}
\end{aligned}$$

The last line comes from bounded second moments in X and from weights summing to 1. The final term can be treated separately for each n.

$$\frac{1}{N_n}\sum_{i_n}\sum_{j_n} W^{(n)}_{i_n,j_n} \left\|\varphi^{(n)}_{i_n} - \varphi^{(n)}_{j_n}\right\|^2$$

Use the triangle inequality,

$$\left\|\varphi_{i_n}^{(n)} - \varphi_{j_n}^{(n)}\right\| \leq \left\|\varphi_{i_n}^{(n)} - \widehat{\varphi}_{i_n}^{(n)}\right\| + \left\|\varphi_{j_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\| + \left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\|$$

and $(a+b+c)^2 = 3(a^2+b^2+c^2)$ to bound this as

$$\frac{1}{N_n}\sum_{i_n}\sum_{j_n} W_{i_n,j_n}^{(n)} O\left(\left\|\varphi_{i_n}^{(n)} - \widehat{\varphi}_{i_n}^{(n)}\right\|^2 + \left\|\varphi_{j_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\|^2 + \left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\|^2\right).$$

Take the first of these terms,

$$\begin{aligned}\frac{1}{N_n}\sum_{i_n}\sum_{j_n} W_{i_n,j_n}^{(n)} \left\|\varphi_{i_n}^{(n)} - \widehat{\varphi}_{i_n}^{(n)}\right\|^2 &= \frac{1}{N_n}\sum_{i_n}\left\|\varphi_{i_n}^{(n)} - \widehat{\varphi}_{i_n}^{(n)}\right\|^2 \sum_{j_n} W_{i_n,j_n}^{(n)} \\ &= O_p(C_n^{-2})\end{aligned}$$

Likewise, the second term is $O_p(C_n^{-2})$ since weights are symmetric and sum to 1.

The third term, with the shorthand $K_{ij,h}^{(n)} = k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\|\right)$, and $\mathbb{1}(a,b)_{ij} = \mathbb{1}\Big\{a < \|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\| < b\Big\}$,

$$\begin{aligned}\frac{1}{N_n}\sum_{i_n}\sum_{j_n} W_{i_n,j_n}^{(n)}\left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\|^2 &= \frac{1}{N_n}\sum_{i_n}\sum_{j_n}\frac{K_{ij,h}^{(n)}\left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\|^2}{\sum_{j_n} K_{ij,h}^{(n)}} \\ &= \frac{1}{N_n}\sum_{i_n}\sum_{j_n}\mathbb{1}(0,h_n)_{ij}\frac{K_{ij,h}^{(n)}\left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\|^2}{\sum_{j_n} K_{ij,h}^{(n)}} \\ &+ \frac{1}{N_n}\sum_{i_n}\sum_{j_n}\sum_{m=1}^{\infty}\mathbb{1}(mh_n,(m+1)h_n)_{ij}\frac{K_{ij,h}^{(n)}\left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\|^2}{\sum_{j_n} K_{ij,h}^{(n)}} \\ &\leq O(h_n^2)\frac{1}{N_n}\sum_{i_n}\frac{\sum_{j_n}\mathbb{1}(0,h_n)_{ij}K_{ij,h}^{(n)}}{\sum_{j_n} K_{ij,h}^{(n)}} \\ &+ \frac{h_n^2}{N_n}\sum_{i_n}\sum_{m=1}^{\infty}(m+1)^2\frac{\sum_{j_n}\mathbb{1}(mh_n,(m+1)h_n)_{ij}K_{ij,h}^{(n)}}{\sum_{j_n} K_{ij,h}^{(n)}}\end{aligned} \tag{A.5}$$

The first term after the last inequality is $O(h_n^2)$ since the denominator always dominates the numerator. For the last term, for each i_n, the term

$$\sum_{m=1}^{\infty}(m+1)^2\sum_{j_n}\mathbb{1}(mh_n,(m+1)h_n)_{ij}K_{ij,h}^{(n)} \leq \sum_{j_n} 2^2 K_{ij,h}^{(n)}$$

because for each i_n, j_n the indicator function can only be 1 for one $m \in \{1,\ldots,\infty\}$, and since $(m+1)^2 K_{ij,h}^{(n)}$ is decreasing in m, from Assumption 5.(iv), across the indicator functions the term is maximised at $m = 1$. Hence, the last line is bounded $O(h_n^2)$.

This makes the whole term bounded $O_p(h_n^2)$. Hence, for each n,

$$\frac{1}{N_n}\sum_{i_n}\sum_{j_n} W_{i_n,j_n}^{(n)}\left\|\varphi_{i_n}^{(n)}-\varphi_{j_n}^{(n)}\right\|^2 \leq \left(O_p\left(C_n^{-2}\right)+O_p\left(h_n^2\right)\right).$$

Taking the product of these terms over all dimensions then forms the statement of the result.

■

Proof of Theorem 3. Begin with (14),

$$\widehat{\beta}_{IC}-\beta^0=\widehat{\Omega}_X^{-1}B+\widehat{\Omega}_X^{-1}\mathrm{vec}_K(\widehat{\boldsymbol{\eta}})'\mathrm{vec}(\boldsymbol{\varepsilon})$$

where, the bias term, $B=B_1+B_2$. Let $N:=\prod_{n=1}N_n$. Then,

$$\begin{aligned} B_1 &:= \mathrm{vec}_K(\widehat{\boldsymbol{\eta}})'\mathrm{vec}(\widehat{\boldsymbol{\Gamma}}_X\cdot(\beta^0-\widetilde{\beta}))\\ B_2 &:= \mathrm{vec}_K(\widehat{\boldsymbol{\eta}})'\mathrm{vec}(\boldsymbol{\mathcal{A}}-\widehat{\boldsymbol{\mathcal{A}}}). \end{aligned}$$

To proceed, firstly, need to show $N^{-1/2}B_1$ and $N^{-1/2}B_2$ are $o_p(1)$.

$$\begin{aligned} \frac{1}{\sqrt{N}}B_1 &= \frac{1}{\sqrt{N}}\mathrm{vec}_K(\boldsymbol{\Gamma}_X-\widehat{\boldsymbol{\Gamma}}_X)'\mathrm{vec}(\widehat{\boldsymbol{\Gamma}}_X\cdot(\beta^0-\widetilde{\beta}))+\frac{1}{\sqrt{N}}\mathrm{vec}_K(\boldsymbol{\eta})'\mathrm{vec}(\widehat{\boldsymbol{\Gamma}}_X\cdot(\beta^0-\widetilde{\beta}))\\ &\leq O_p(\sqrt{N}\xi_X\xi_\beta)\iota_K\sum_{k=1}^{K}\frac{1}{\sqrt{N}}\left\|\widehat{\boldsymbol{\Gamma}}_{X_k}\right\|_F+O_p(\xi_\beta)\frac{1}{\sqrt{N}}\sum_{k=1}^{K}\mathrm{vec}_K(\boldsymbol{\eta})'\mathrm{vec}(\widehat{\boldsymbol{\Gamma}}_{X_k}). \end{aligned}$$

Boundedness of $N^{-1/2}\left\|\widehat{\boldsymbol{\Gamma}}_{X_k}\right\|_F$ comes from Assumption 1.(i). From Assumption 11, the first term is then $o_p(1)$. Assumptions 11- 12 bound the second term as $O_p(\xi_\beta)=o_p(1)$.

$$\frac{1}{\sqrt{N}}B_2=O_p(\sqrt{N}\xi_X\xi_{\mathcal{A}})\iota_K+\frac{1}{\sqrt{N}}\mathrm{vec}_K(\boldsymbol{\eta})'\mathrm{vec}(\boldsymbol{\mathcal{A}}-\widehat{\boldsymbol{\mathcal{A}}}).$$

The first term is $o_p(1)$ by Assumption 11. For the second term,

$$\mathrm{var}\left(\frac{1}{\sqrt{N}}\mathrm{vec}_K(\boldsymbol{\eta})'\mathrm{vec}(\boldsymbol{\mathcal{A}}-\widehat{\boldsymbol{\mathcal{A}}})\right)=\mathbb{E}(\eta_{i_1,\ldots,i_d}\eta'_{i_1,\ldots,i_d})\frac{1}{N}\sum_{i_1,\ldots,i_d}(\mathcal{A}_{i_1,\ldots,i_d}-\widehat{\mathcal{A}}_{i_1,\ldots,i_d})^2$$

where the equality comes from i.i.d. $\eta_{i_1,\ldots,i_d}$ and Assumption 12. This term is $o_p(1)$ from Assumption 11. Also, $\mathbb{E}\left(N^{-1/2}\mathrm{vec}_K(\boldsymbol{\eta})'\mathrm{vec}(\boldsymbol{\mathcal{A}}-\widehat{\boldsymbol{\mathcal{A}}})\right)=0$. Hence, $N^{-1/2}B_2=o_p(1)$.

Next,

$$\frac{1}{\sqrt{N}}\mathrm{vec}_K(\widehat{\boldsymbol{\eta}})'\mathrm{vec}(\boldsymbol{\varepsilon})=\frac{1}{\sqrt{N}}\mathrm{vec}_K(\boldsymbol{\Gamma}_X-\widehat{\boldsymbol{\Gamma}}_X)'\mathrm{vec}(\boldsymbol{\varepsilon})+\frac{1}{\sqrt{N}}\mathrm{vec}_K(\boldsymbol{\eta})'\mathrm{vec}(\boldsymbol{\varepsilon}).$$

The second term is asymptotically normally distributed by Assumption 16. The first term is mean zero and has variance,

$$\begin{aligned} \mathbb{E}\Bigg[&\frac{1}{N}\sum_{i_1,\ldots,i_d}\left(\boldsymbol{\Gamma}_{X,i_1,\ldots,i_d}-\widehat{\boldsymbol{\Gamma}}_{X,i_1,\ldots,i_d}\right)\left(\boldsymbol{\Gamma}_{X,i_1,\ldots,i_d}-\widehat{\boldsymbol{\Gamma}}_{X,i_1,\ldots,i_d}\right)'\varepsilon^2_{i_1,\ldots,i_d}+\ldots\\ &\cdots+\frac{1}{N}\sum_{i'_1\ldots i'_d\neq i_1\ldots i_d}\sum_{i_1,\ldots,i_d}\left(\boldsymbol{\Gamma}_{X,i_1,\ldots,i_d}-\widehat{\boldsymbol{\Gamma}}_{X,i_1,\ldots,i_d}\right)\left(\boldsymbol{\Gamma}_{X,i'_1,\ldots,i'_d}-\widehat{\boldsymbol{\Gamma}}_{X,i'_1,\ldots,i'_d}\right)'\varepsilon_{i_1,\ldots,i_d}\varepsilon_{i'_1,\ldots,i'_d}\Bigg] \end{aligned}$$

The first term is $o_p(1)$ from Assumption 11- 12 and bounded $\mathbb{E}\varepsilon^2_{i_1,\ldots,i_d}$. For the second term use shorthand index notation i in place of $i_1,\ldots,i_d$. From Assumption 12, for each k,k',

$$
\begin{aligned}
&\frac{1}{N}\sum_{j\neq i}\sum_{i}\mathbb{E}\left[\left(\mathbf{\Gamma}_{X_k,i}-\widehat{\mathbf{\Gamma}}_{X_k,i}\right)\left(\mathbf{\Gamma}_{X_{k'},j}-\widehat{\mathbf{\Gamma}}_{X_{k'},j}\right)\right]\mathbb{E}\left[\varepsilon_i\varepsilon_j\right] \\
&\qquad\leq\left(\frac{1}{N}\sum_{j\neq i}\sum_{i}\mathbb{E}\left[\left(\mathbf{\Gamma}_{X_k,i}-\widehat{\mathbf{\Gamma}}_{X_k,i}\right)\left(\mathbf{\Gamma}_{X_{k'},j}-\widehat{\mathbf{\Gamma}}_{X_{k'},j}\right)\right]^2\right)^{1/2}\left(\frac{1}{N}\sum_{j\neq i}\sum_{i}\mathbb{E}\left[\varepsilon_i\varepsilon_j\right]^2\right)^{1/2} \\
&\qquad\leq\left(\frac{1}{N}\sum_{j\neq i}\sum_{i}\mathbb{E}\left[\left(\mathbf{\Gamma}_{X_k,i}-\widehat{\mathbf{\Gamma}}_{X_k,i}\right)^2\right]\mathbb{E}\left[\left(\mathbf{\Gamma}_{X_{k'},j}-\widehat{\mathbf{\Gamma}}_{X_{k'},j}\right)^2\right]\right)^{1/2}\left(\frac{1}{N}\sum_{j\neq i}\sum_{i}\sigma^2_{ij}\right)^{1/2} \\
&\qquad= O_p\left(N^{1/2}\xi_X^2\xi_{\sigma,N}\right).
\end{aligned}
$$

Notation comes from Assumption 17, which also implies $O_p\left(N^{1/2}\xi_X^2\xi_{\sigma,N}\right)=o_p(1)$.

Finally, $\widehat{\Omega}_X^{-1}=(\mathbb{E}(\eta_{i_1,\ldots,i_d}\eta'_{i_1,\ldots,i_d})+o_p(1))^{-1}$. Assumption 16 gives the asymptotic distribution for $N^{-1/2}\text{vec}_K(\boldsymbol{\eta})'\text{vec}(\boldsymbol{\varepsilon})$, which completes the proof.

■

A.3 Iterative Estimator

Assumption 6 can be significantly relaxed with the use of an iterative projection of fixed-effects. The following estimation procedure allows for a higher dimensional set of proxy measures used to form projection weights. Namely, the sampling restriction can be entry-wise, i.e. on scalars, rather than on the neighbourhoods of entire vectors. The procedure is general, and can use many weights, or even clusters, in place of the kernel weights used to describe the estimator below.

Every tensor has a higher-order singular value decomposition (HOSVD),

$$\mathcal{A}=\mathcal{S}\times(U^{(1)},\ldots,U^{(d)})$$

where $\mathcal{S}\in\mathbb{R}^{\times_{n=1}^d r_n}$ and $U^{(n)}\in\mathbb{R}^{N_n\times r_n}$, where r_n is the n-th compononent of the multilinear rank. When written in the above form where $U^{(n)}\in\mathbb{R}^{N_n\times r_n}$ rather than $U^{(n)}\in\mathbb{R}^{N_n\times N_n}$, this can be referred to as the compact HOSVD. The iterative estimator works as follows.

1. Obtain proxies for $U^{(n)}$, called $\widehat{U}^{(n)}$ for each $n=1,\ldots,d$. Use these to form weights, for $m_n=1,\ldots,r_n$;

$$W_{nm_n,i_nj_n}:=\frac{k\left(\frac{1}{h_n}\left(\widehat{U}^{(n)}_{i_n,m_n}-\widehat{U}^{(n)}_{j_n,m_n}\right)^2\right)}{\sum_{i'_n=1}^{N_n}k\left(\frac{1}{h_n}\left(\widehat{U}^{(n)}_{i_n,m_n}-\widehat{U}^{(n)}_{i'_n,m_n}\right)^2\right)}\qquad\text{for each } n=1,\ldots,d.\tag{A.6}$$

2. For each $m_n = 1, \ldots, r_n$ project out weighted means from $\boldsymbol{Y}$ and $\boldsymbol{X}_k$ for each $k = 1, \ldots, K$ along each dimension, $n = 1, \ldots, d$ as follows,

$$\widetilde{\mathcal{A}} = \mathcal{A} \times \left(\prod_{m_1=1}^{r_1} \left(\mathbb{I} - W_{1m_1}\right), \ldots, \prod_{m_d=1}^{r_d} \left(\mathbb{I} - W_{dm_d}\right) \right) \tag{A.7}$$

3. Perform pooled OLS of $\widetilde{\boldsymbol{Y}}$ on $\widetilde{\boldsymbol{X}}$. Call the estimator $\widehat{\beta}_{IK}$, for iterative kernel.

From the representation in (A.7) the proof techniques from previous results follow straightforwardly. As is proposed in the main text, proxies $\widehat{U}^{(n)}$ can be estimated using the matrix method with each dimension used as rows. Then results from Proposition A.1 in Bai (2009) can be used to bound the estimation error of these proxies with respect to the true $U^{(n)}$. Greater care is needed for the fact Proposition A.1 in Bai (2009) is an up-to-rotations result that is not required in previous results, hence this is taken care of specifically in the proof.

Assumption A.1 (Regularity of proxy measures)**.** *For $0 < \rho \leq 1$ Define $M_n^h\left(\widehat{U}_{i_n,m_n}^{(n)}\right)$ as*

$$M_n^h\left(\widehat{U}_{i_n,m_n}^{(n)}\right) := \frac{1}{N_n^\rho} \sum_{j_n=1}^{N_n} \mathbb{1}\left(\widehat{U}_{j_n,m_n}^{(n)} \in B_{he}\left(\widehat{U}_{i_n,m_n}^{(n)}\right)\right),$$

where $B_{he}(x)$ is an he-neighbourhood around x. Then for each dimension n, for any $e > 0$, as $h \to 0$ there exists $0 < \rho \leq 1$ such that $N_n h \to \infty$,

$$\frac{1}{N_n} \sum_{i_n=1}^{N_n} M_n^h\left(\widehat{U}_{i_n,m_n}^{(n)}\right) > 0 \qquad \textit{wpa. } 1.$$

Proposition A.1 (Upper bound on iterative estimator)**.** *Let Assumption 5, Assumption 7 specifically for the iterative transformation, and Assumption A.1 hold. For the higher order singular value decomposition of the fixed-effects term, let $\frac{1}{N_{n^*}} \sum_{i_{n^*}} \left\| U_{i_{n^*}}^{(n^*)} - H^{(n^*)} \widehat{U}_{i_{n^*}}^{(n^*)} \right\|^2 = O_p(C_{n^*}^{-2})$ for $n^* \in \mathcal{M}'$ with $C_{n^*}^{-2} \to 0$, and $\frac{1}{N_{n'}} \sum_{i_{n'}} \left\| U_{i_{n'}}^{(n')} - H^{(n')} \widehat{U}_{i_{n'}}^{(n')} \right\|^2 = O_p(1)$ for $n' \notin \mathcal{M}'$, where $\mathcal{M}'$ is a non-empty subset of dimensions. $H^{(n)}$ is an invertible $r_n \times r_n$ matrix. Let h_n be the bandwidth parameter from Assumption A.1. Then,*

$$\left\| \widehat{\beta}_{IK,\mathcal{W}} - \beta^0 \right\| = O_p\Big(\prod_n r_n^{1/2} \big(C_n^{-1} + r_n^{1/2} h_n\big) \Big) + O_p\left(\prod_{n=1}^d \frac{1}{\sqrt{N_n}} \right).$$

For $h_n \lesssim O(C_n^{-1} r_n^{-1/2})$ this reduces to

$$\left\| \widehat{\beta}_{IK,\mathcal{W}} - \beta^0 \right\| = \prod_n r_n^{1/2} O_p\left(\prod_{n^* \in \mathcal{M}'} O_p\left(C_{n^*}^{-1}\right) \right) + O_p\left(\prod_{n=1}^d \frac{1}{\sqrt{N_n}} \right).$$

Proof of Proposition A.1.

$$\frac{1}{N}\sum_{i_1,\dots,i_d}\widetilde{X}_{i_1,\dots,i_d}\sum_{m_1}^{r_1}\cdots\sum_{m_d}^{r_d}\mathcal{S}_{m_1,\dots,m_d}\prod_{m_1'=1}^{r_1}\left(U^{(1)}_{i_1,m_1}-W'_{1m_1',i_1}U^{(1)}_{\cdot,m_1}\right)\cdots\prod_{m_d'=1}^{r_d}\left(U^{(d)}_{i_d,m_d}-W'_{dm_d',i_d}U^{(d)}_{\cdot,m_d}\right)$$

$$=\frac{1}{N}\sum_{i_1,\dots,i_d}\sum_{j_1\dots,j_d}\widetilde{X}_{i_1,\dots,i_d}\sum_{m_1}^{r_1}\cdots\sum_{m_d}^{r_d}\mathcal{S}_{m_1,\dots,m_d}\prod_{m_1'=1}^{r_1}W_{1m_1',i_1j_1}\left(U^{(1)}_{i_1,m_1}-U^{(1)}_{j_1,m_1}\right)\dots$$

$$=\frac{1}{N}\sum_{i_1,\dots,i_d}\sum_{j_1\dots,j_d}\widetilde{X}_{i_1,\dots,i_d}\prod_{m_1'=1}^{r_1}\sqrt{W_{1m_1',i_1j_1}}\cdots\sum_{m_1}^{r_1}\cdots\sum_{m_d}^{r_d}\mathcal{S}_{m_1,\dots,m_d}\prod_{m_1'=1}^{r_1}\sqrt{W_{1m_1',i_1j_1}}\left(U^{(1)}_{i_1,m_1}-U^{(1)}_{j_1,m_1}\right)\dots$$

By Cauchy-Schwarz, this can be bound by

$$\left(\frac{1}{N}\sum_{i_1,\dots,i_d}\sum_{m_1}^{r_1}\cdots\sum_{m_d}^{r_d}\widetilde{X}^2_{i_1,\dots,i_d}\sum_{j_1\dots,j_d}\prod_{m_1'=1}^{r_1}W_{1m_1',i_1j_1}\cdots\prod_{m_d'=1}^{r_d}W_{dm_d',i_dj_d}\right)^{1/2}\dots \tag{A.8}$$

$$\dots\left(\frac{1}{N}\sum_{i_1,\dots,i_d}\sum_{j_1\dots,j_d}\sum_{m_1}^{r_1}\cdots\sum_{m_d}^{r_d}\mathcal{S}^2_{m_1,\dots,m_d}\prod_{m_1'=1}^{r_1}W_{1m_1',i_1j_1}\left(U^{(1)}_{i_1,m_1}-U^{(1)}_{j_1,m_1}\right)^2\dots\right)^{1/2}$$

Focus on the first term in (A.8) and note

$$\sum_{j_1\dots,j_d}\prod_{m_1'=1}^{r_1}W_{1m_1',i_1j_1}\cdots\prod_{m_d'=1}^{r_d}W_{dm_d',i_dj_d}=\sum_{j_1}\prod_{m_1'=1}^{r_1}W_{1m_1',i_1j_1}\cdots\sum_{j_d}\prod_{m_d'=1}^{r_d}W_{dm_d',i_dj_d}$$

such that for any n,

$$\begin{aligned}
\sum_{j_n}\prod_{m_n'=1}^{r_1}W_{nm_n',i_nj_n} &\leq \left(\sum_{j_n}W^2_{n1,i_nj_n}\right)^{1/2}\left(\sum_{j_n}\prod_{m_n'\neq 1}W^2_{nm_n',i_nj_n}\right)^{1/2}\\
\text{(weights in [0,1])}\qquad &\leq \left(\sum_{j_n}W_{n1,i_nj_n}\right)^{1/2}\left(\sum_{j_n}\prod_{m_n'\neq 1}W_{nm_n',i_nj_n}\right)^{1/2}\\
\text{(weights sum to 1)}\qquad &\leq 1\cdot\left(\sum_{j_n}W^2_{n2,i_nj_n}\right)^{1/2}\left(\sum_{j_n}\prod_{m_n'\neq\{1,2\}}W^2_{nm_n',i_nj_n}\right)^{1/2}\dots
\end{aligned}$$

which is subsequently bound by 1. By bounded second moments of X post transformation, this bounds the first term as $O_p\left(\prod_n r_n^{1/2}\right)$.

Now for the second term in (A.8). For singular values, $\mathcal{S}_{m_1,\dots,m_d}$, bounded in probability this term is bounded as,

$$\begin{aligned}
&\max_{m_1,\dots,m_d}\mathcal{S}^2_{m_1,\dots,m_d}\prod_{n=1}^{d}\sum_{m_n}^{r_n}\frac{1}{N_n}\sum_{i_n}\sum_{j_n}\prod_{m_n'=1}^{r_n}W_{nm_n',i_nj_n}\left(U^{(n)}_{i_n,m_n}-U^{(n)}_{j_n,m_n}\right)^2\\
&\qquad=O_p(1)\prod_{n=1}^{d}\sum_{m_n}^{r_n}\frac{1}{N_n}\sum_{i_n}\sum_{j_n}\prod_{m_n'=1}^{r_n}W_{nm_n',i_nj_n}\left(U^{(n)}_{i_n,m_n}-U^{(n)}_{j_n,m_n}\right)^2
\end{aligned}$$

Focus for each n, where the superscript notation in $H^{(n)}$ is suppressed for brevity,

$$
\begin{aligned}
\sum_{m_n}^{r_n} \frac{1}{N_n} \sum_{i_n,j_n} \prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n} \big(U^{(n)}_{i_n,m_n} - U^{(n)}_{j_n,m_n}\big)^2 &\leq \dots \\
\dots \leq \sum_{m_n}^{r_n} \frac{3}{N_n} \sum_{i_n,j_n} \prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n} \big(U^{(n)}_{i_n,m_n} - H'_{m_n} \widehat{U}^{(n)}_{i_n}\big)^2 & \\
+ \sum_{m_n}^{r_n} \frac{3}{N_n} \sum_{i_n,j_n} \prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n} \big(U^{(n)}_{j_n,m_n} - H'_{m_n} \widehat{U}^{(n)}_{j_n}\big)^2 & \\
+ \sum_{m_n}^{r_n} \frac{3}{N_n} \sum_{i_n,j_n} \prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n} \big(H'_{m_n} \widehat{U}^{(n)}_{i_n} - H'_{m_n} \widehat{U}^{(n)}_{j_n}\big)^2 &
\end{aligned}
$$

by Jensen's inequality. The first term,

$$
\begin{aligned}
\frac{3}{N_n} \sum_{i_n} \sum_{m_n}^{r_n} \big(U^{(n)}_{i_n,m_n} - H'_{m_n} \widehat{U}^{(n)}_{i_n}\big)^2 \sum_{j_n} \prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n} &\leq \frac{3}{N_n} \sum_{i_n} \big\| U^{(n)}_{i_n} - H' \widehat{U}^{(n)}_{i_n} \big\|^2 \cdot 1 \\
&= O_p(C_n^{-2})
\end{aligned}
$$

Likewise, the second term is $O_p(C_n^{-2})$.

The final term,

$$
\begin{aligned}
\frac{3}{N_n} \sum_{i_n,j_n} \prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n} \sum_{m_n}^{r_n} \Big(H'_{m_n} \big(\widehat{U}^{(n)}_{i_n} - \widehat{U}^{(n)}_{j_n}\big)\Big)^2 &\leq \dots \\
\dots \leq \frac{3}{N_n} \sum_{i_n,j_n} \prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n} \big\| \widehat{U}^{(n)}_{i_n} - \widehat{U}^{(n)}_{j_n} \big\|^2 \sum_{m_n}^{r_n} \| H_{m_n} \|^2 & \\
= O_p(r_n) \frac{1}{N_n} \sum_{i_n,j_n} \sum_{m_n}^{r_n} \prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n} \big(\widehat{U}^{(n)}_{i_nm_n} - \widehat{U}^{(n)}_{j_nm_n}\big)^2, &
\end{aligned}
$$

where the last bound results from H being invertible, hence has bounded singular values. Since $\sum_{m_n}^{r_n} \|H_{m_n}\|^2 = \sum_{m_n}^{r_n} \sum_{m'_n}^{r_n} H^2_{m_n,m'_n} = \sum_{r=1}^{r_n} \sigma_r^2(H)$, where $\sigma_r^2(H)$ are the bounded singular values of H (bounded from invertibility of H). Without loss, assume the ordering of m_n and m'_n from the sum and product in this last expression are the same. Then, for each m_n

$$
\begin{aligned}
\frac{1}{N_n} \sum_{i_n,j_n} \prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n} \big(\widehat{U}^{(n)}_{i_nm_n} - \widehat{U}^{(n)}_{j_nm_n}\big)^2 &= \frac{1}{N_n} \sum_{i_n,j_n} W_{nm_n,i_nj_n} \big(\widehat{U}^{(n)}_{i_nm_n} - \widehat{U}^{(n)}_{j_nm_n}\big)^2 \prod_{m'_n \neq m_n} W_{nm'_n,i_nj_n} \\
&\leq \frac{1}{N_n} \sum_{i_n,j_n} W_{nm_n,i_nj_n} \big(\widehat{U}^{(n)}_{i_nm_n} - \widehat{U}^{(n)}_{j_nm_n}\big)^2 \cdot 1
\end{aligned}
$$

As in the proof of Proposition 2, this final term is bound as $O_p(h_n^2)$, hence, the second term of (A.8) is $O_p(r_n h_n^2)$.

Thus, consistency is at the rate $O_p\Big(\prod_n r_n^{1/2}(C_n^{-1} + r_n^{1/2} h_n)\Big)$.

■